

The image shows a well-lit industrial networking lab. In the foreground, there are two long workbenches with light-colored wooden tops and black frames containing multiple rows of silver toolboxes. A black ergonomic chair is positioned at one of the workbenches. In the background, several racks of industrial networking equipment are visible, each with various cables and components. A projector is mounted on the ceiling, and the room has a clean, professional appearance with a grey floor and a white grid ceiling with recessed lighting.

INDUSTRIAL NETWORKING

MENG 4010

CONTENTS

5. TCP/IP Protocols

- 5.1. TCP/IP Protocols – Overview
- 5.2. Application Layer Protocols
- 5.3. Transport Layer Protocols
- 5.4. Internetwork Layer Protocols

5.1 TCP/IP PROTOCOLS - OVERVIEW

- General meaning: A protocol is a set of rules or guidelines for communication.
 - Example (human): Two people must share a common language (English, Spanish, etc.) to understand each other.
- Past:
 - Computers in the 1990s used different protocols:
 - NetBEUI → for Windows networks.
 - IPX/SPX → for Novell networks.
 - These are obsolete now.
- Today: Everyone uses TCP/IP, the protocol of the Internet.

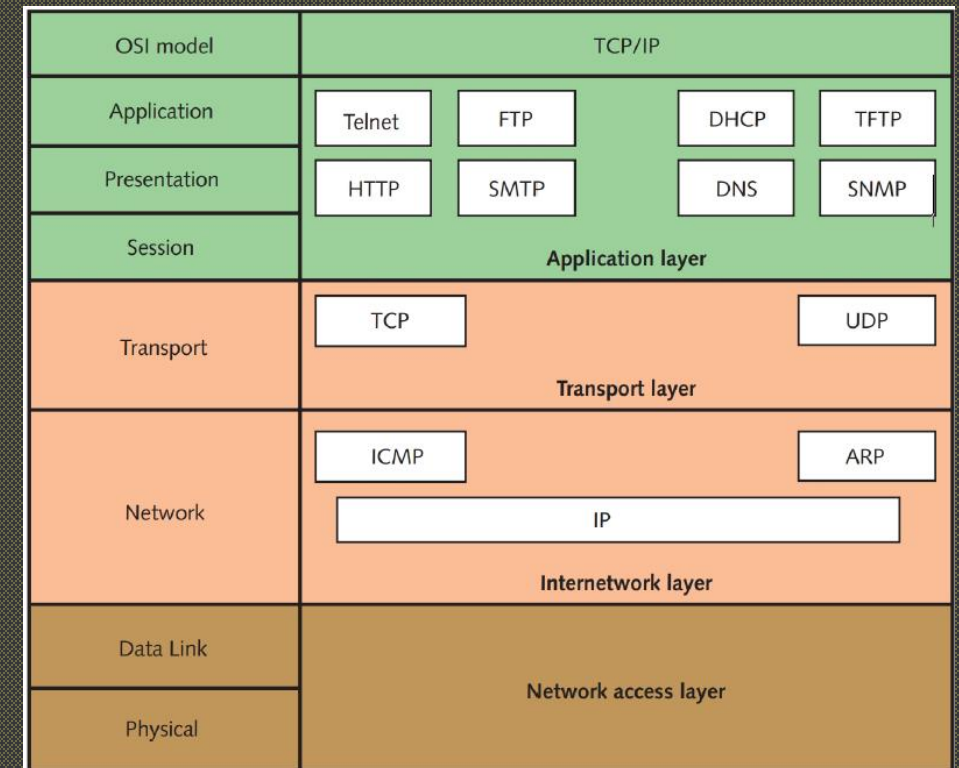
TCP/IP - Transmission Control Protocol/Internet Protocol

- A protocol suite = a group of protocols working together.
- The most common one = TCP/IP protocol suite, which has more than a dozen protocols organized in layers.

NetBEUI - NetBIOS Extended User Interface

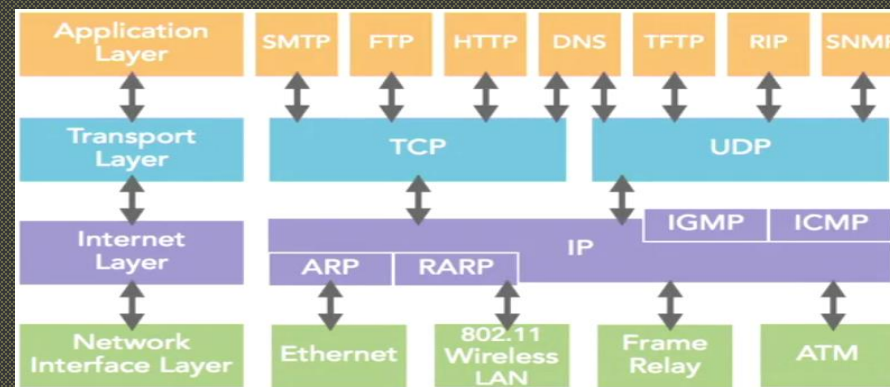
NetBIOS - Network Basic Input/Output System

IPX/SPX - Internetwork Packet Exchange/Sequenced Packet Exchange



5.1 TCP/IP PROTOCOLS - OVERVIEW

- Application Layer → User-facing services.
 - Protocols: HTTP (web), FTP (file transfer), SMTP (email), DNS (domain lookup).
 - Example: When you type www.humber.ca, your browser uses HTTP to request the page.
- Transport Layer → Ensures data is sent correctly.
 - Protocols: TCP (reliable, like a registered mail) or UDP (faster, like regular mail).
 - Example: HTTP uses TCP, so no part of the webpage is lost.
- Internetwork Layer → Moves data between devices across networks.
 - Protocols: IP (IPv4, IPv6), ICMP (error messages), ARP (finding MAC addresses).
 - Example: Your request packet is given an IP address to reach the humber.ca server.
- Network Access Layer → Physical delivery of data over cables, Wi-Fi, etc.
 - Protocols: Ethernet, Wi-Fi, WAN technologies.
 - Example: The packet is converted into bits and sent through your Wi-Fi router.



https://www.uomus.edu.iq/img/lectures21/MUCLecture_2025_2218449.pdf

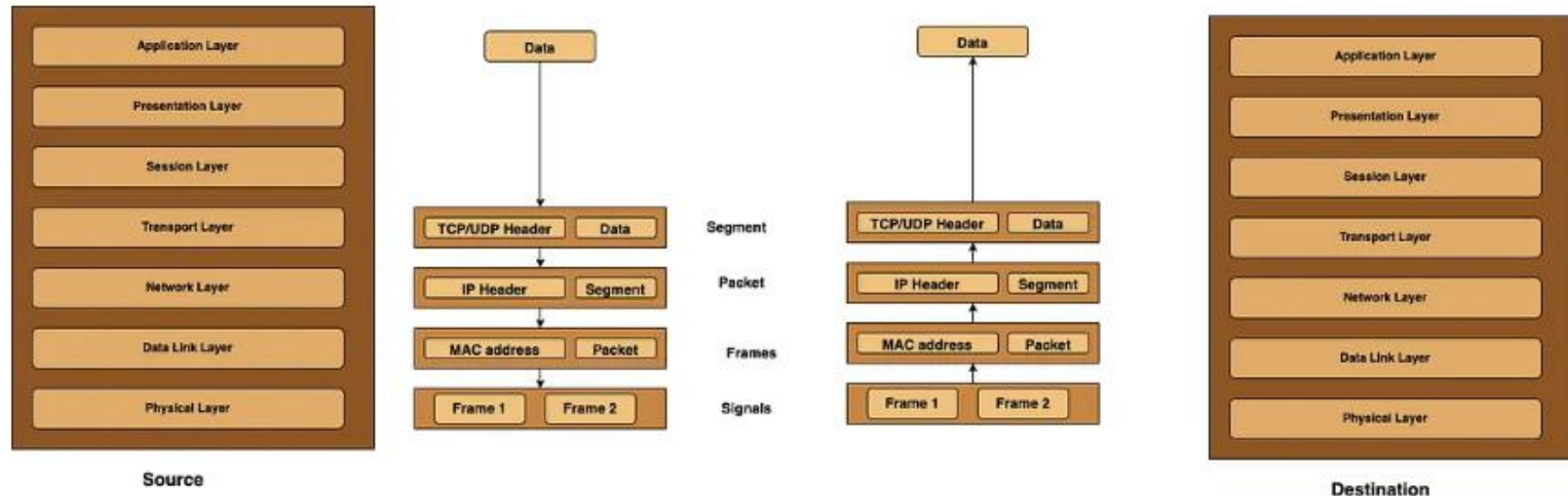
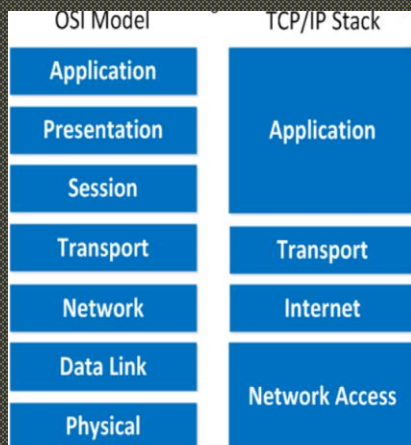
5.1 TCP/IP PROTOCOLS - OVERVIEW

- How the Layers Work Together - Example: visiting a website
- You open your browser and request `www.humber.ca`.
 - Application Layer: HTTP creates the data request.
 - Transport Layer (TCP): Adds a TCP header → now called a segment.
 - Internetwork Layer (IP): Adds an IP header → now called a packet.
 - Network Access Layer: Adds a frame header and trailer → now called a frame.
 - Frame is sent as bits over the network to the server.
- The server processes it and sends the webpage back.

Sending a webpage request is like mailing a package:

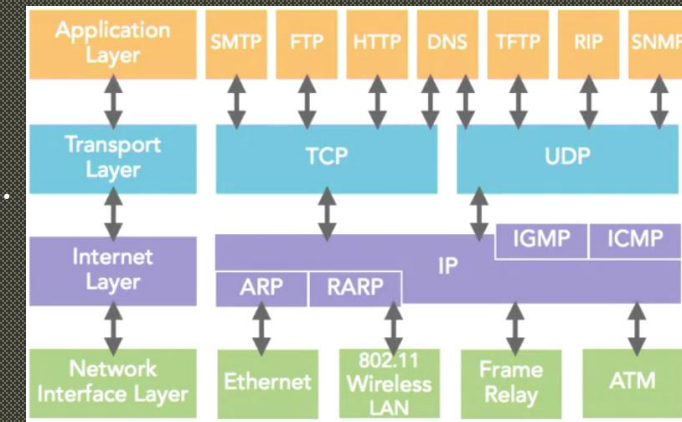
- Application Layer: Writes the letter (data).
- Transport Layer: Puts it in an envelope with tracking info (segment).
- Internetwork Layer: Adds the address and postal code (packet).
- Network Access Layer: Hands it to the post office, which wraps and delivers it (frame → bits).

Frame header IP header TCP header get the humber.ca home page Frame trailer



5.2 APPLICATION LAYER PROTOCOLS

- Provides network services to user applications (like Word, Web browsers, or email clients).
 - Makes it possible for applications to access network resources (files, printers, web pages, etc.).
 - Examples of Application Layer Protocols
 - File Sharing:
 - Windows: SMB/CIFS (Server Message Block / Common Internet File System).
 - Linux: NFS and Samba.
 - Example: Opening a file on a network server through Microsoft Word.
 - Web & Email:
 - HTTP/HTTPS → Browsers requesting web pages.
 - SMTP → Sending/receiving emails.
 - Example: HTTPS adds encryption & authentication when logging into secure sites (like online banking).
 - Network Services:
 - DNS → Converts website names (google.com) into IP addresses.
 - DHCP → Automatically assigns IP addresses.
- SMB/CIFS (Windows): Enables file and folder sharing over a network for Windows.
 - Example: Opening a Word document from a shared office server.
 - NFS (Linux): Network File System - Allows file sharing across Linux systems.
 - Example: Accessing a shared folder on a server to edit a file.
 - Samba (Linux/Windows): Facilitates file sharing between Linux and Windows.
 - Example: A Linux system accessing a Windows shared folder.



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5.2 APPLICATION LAYER PROTOCOLS

How It Works

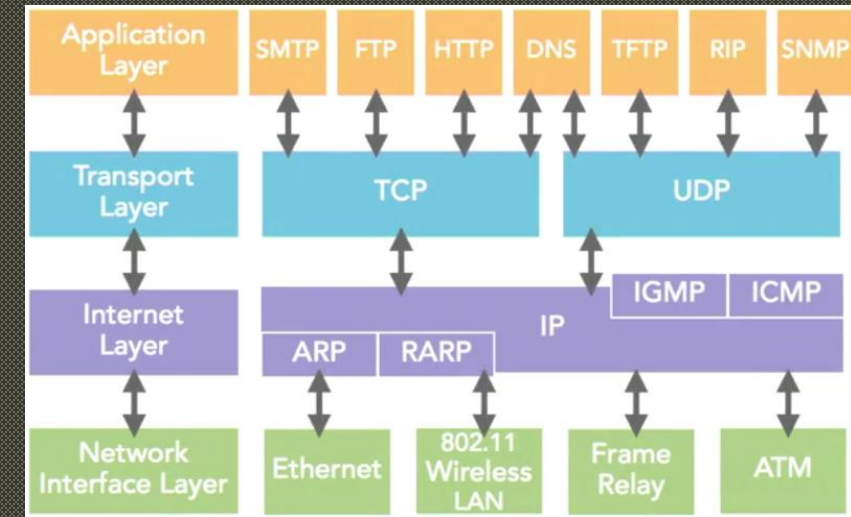
- Application (e.g., Web browser) prepares the data request.
- Application Layer protocol (e.g., HTTP) formats the request.
- Sends it down the protocol stack to the Transport Layer, then to lower layers until the data becomes bits sent on the network.

Extra Services by Application Layer

- Authentication → Manages logins/passwords.
 - Example: Entering a username/password when accessing a shared folder.
- Data Formatting & Translation → Encrypts, compresses, or converts data.
 - Example: HTTPS encrypts traffic for secure communication.

Client-Server Relationship

- Most Application-layer protocols have two sides:
 - Client side: e.g., Web browser, email client, Word.
 - Server side: e.g., Apache Web Server, Microsoft IIS, Mail Server.
- Example:
 - Client (browser) sends a request using HTTP.
 - Web Server Software (Apache or IIS) responds with the web page.



https://www.uomus.edu.iq/img/lectures21/MUCLecture_2025_2218449.pdf

5.2 APPLICATION LAYER PROTOCOLS

HTTP – PROTOCOL OF WORLD WIDE WEB

- HTTP (Hypertext Transfer Protocol) = The main protocol of the World Wide Web.
- Used by Web browsers to request and display content from web servers.
- Designed to transfer static web pages written in HTML.

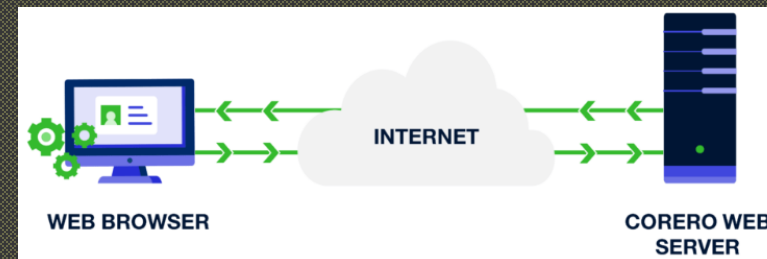
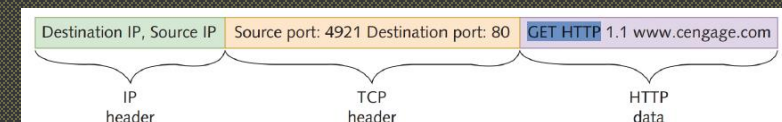
How HTTP Works

- The browser (client) sends a request to the server.
 - Example: GET /index.html HTTP/1.1
 - Hello server, please send me the file /index.html using the rules of HTTP/1.1
- The server responds with the requested page or resource.

Transport Layer Details

- Uses TCP to ensure reliable delivery.
- Default port: 80.
 - Example: Typing `http://www.humber.ca` → the browser connects on TCP port 80.

- GET is the HTTP method, meaning the client is asking the server to send back (retrieve) a resource, usually a web page, image, or file.
- /index.html is the path of the resource being requested on the server. If the server is hosting `www.cengage.com`, then the full URL would be:
`http://www.cengage.com/index.html`
- HTTP/1.1 is the HTTP version being used, which requires the client to send a Host header (e.g., `Host: www.cengage.com`) so the server knows which website to serve if it's hosting multiple sites.



<https://www.corero.com/what-is-http-and-how-does-it-work/>

5.2 APPLICATION LAYER PROTOCOLS

EMAIL PROTOCOLS

POP3 (Post Office Protocol v3)

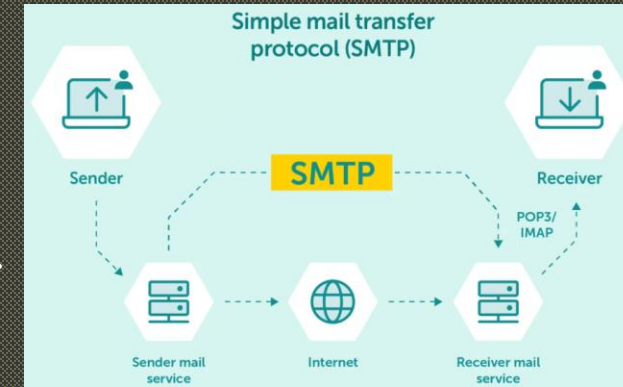
- Used by email clients to download messages from the server to the local computer.
- Emails are removed from the server once downloaded (like taking mail out of a mailbox).
- Port: TCP 110.
- Example: Using Outlook to download all your emails so they are stored only on your laptop.

IMAP4 (Internet Message Access Protocol v4)

- Allows emails to be viewed and managed directly on the server.
 - Downloads headers first; the full message and attachments are downloaded when opened.
 - Messages usually remain on the server → you can access them from multiple devices.
- Port: TCP 143.
- Example: Checking Gmail on both your phone and laptop, and messages stay synced.

SMTP (Simple Mail Transfer Protocol)

- Standard protocol for sending emails.
- Works with POP3 or IMAP (which are for receiving emails).
- Port: TCP 25.
- Example: When you hit “Send” in Outlook or Gmail, SMTP delivers the message to the recipient’s mail server.



<https://emaillabs.io/en/email-protocols-learn-the-difference-between-imap-pop3-and-smtp/>

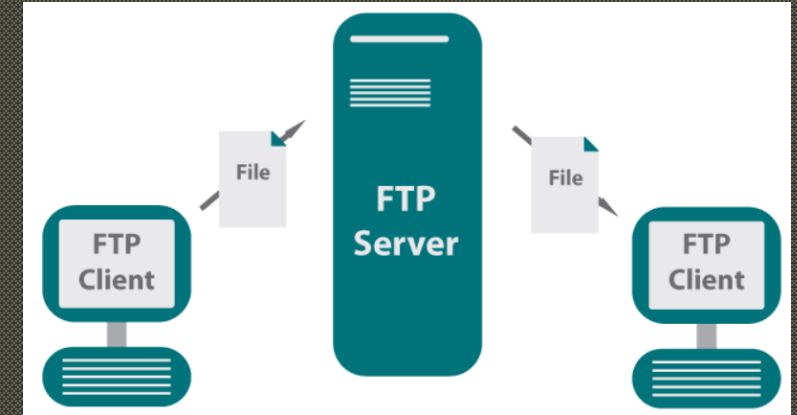
- POP3 → Like picking up all letters from your mailbox and keeping them at home (they are no longer in the mailbox).
- IMAP → Like leaving letters in the mailbox but reading copies whenever you want, from any device.
- SMTP → Like the postal service that actually sends your letters out.

5.2 APPLICATION LAYER PROTOCOLS

FTP PROTOCOL

FTP (File Transfer Protocol)

- Transfers and manages files between client ↔ server.
- Ports (using TCP Protocol):
 - 21 → Control commands (login, navigation).
 - 20 → Actual file data transfer.
- Features:
 - Supports file management (upload, download, rename, delete).
 - Works on both private networks and the Internet.
- Security issue:
 - Not secure (username, password, and data are sent in plain text).
 - Anyone intercepting packets can see usernames, passwords, and files.
 - Security only comes with FTPS (FTP over SSL/TLS) or SFTP (SSH File Transfer Protocol, which is actually a different protocol).
- Example: Uploading a website's HTML files to a hosting server.
- Like a delivery service with full options (tracking, ID check, package organization)



<https://www.progress.com/blogs/what-is-file-transfer-protocol-ftp>

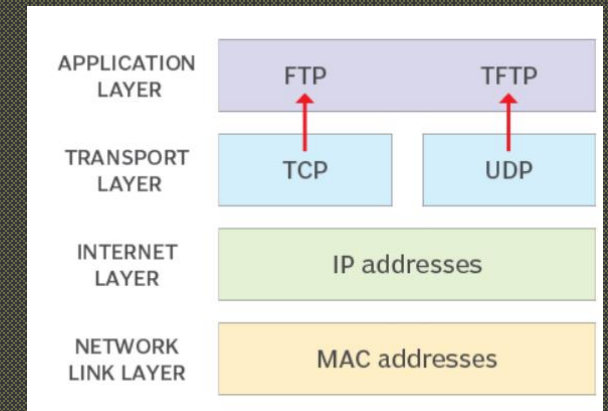
SSL - Secure Sockets Layer
TLS - Transport Layer Security

5.2 APPLICATION LAYER PROTOCOLS

TFTP PROTOCOL

TFTP (Trivial File Transfer Protocol)

- A simpler, lightweight file transfer protocol.
 - Port: 69 (uses UDP).
 - Features:
 - Minimal functionality → only transfers files (no file management like rename or delete).
 - Less reliable (UDP = no guaranteed delivery).
 - Typical use:
 - Inside LANs for small transfers.
 - Uploading firmware/configuration to routers/switches.
 - Network booting (devices loading their OS from a server instead of a hard drive).
 - Security issue:
 - No encryption or login security.
 - But less risky since it is usually used only in closed, internal networks.
 - Example: A router downloading its startup configuration file from a TFTP server.
-
- FTP → Large, secure, managed file transfers (e.g., websites, cloud storage).
 - TFTP → Small, quick, internal transfers (e.g., router firmware update).



<https://www.techtarget.com/searchnetworking/tutorial/How-to-use-TFTP-to-transfer-files-across-the-network>

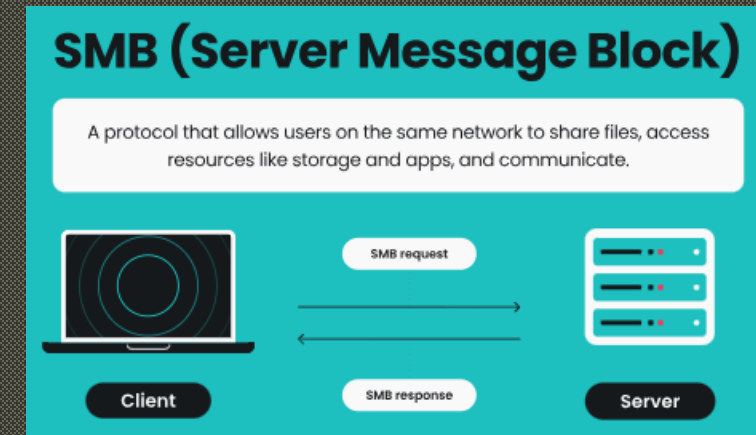
5.2 APPLICATION LAYER PROTOCOLS

SMB PROTOCOL

- SMB (Server Message Block) is a protocol for file and printer sharing in networks.
- Used mostly by Windows systems, but also supported by Linux (Samba) and Mac OS X.

Features

- Resource Sharing: Lets computers share files, folders, and printers over a network.
- Client/Server Setup:
 - Client for Microsoft Networks → the requester.
 - File and Printer Sharing for Microsoft Networks → the provider.
- Port: Uses TCP port 445.
- Scope: Works mainly in private/local networks (LANs), not across the Internet.



<https://surfsark.com/blog/what-is-smb?>

Example

- On a Windows network:
 - Computer A enables “File and Printer Sharing.”
 - Computer B (on the same LAN) accesses Computer A’s shared folder via SMB.
- On Linux:
 - Samba software allows Linux machines to share files with Windows PCs using SMB.

SMB is like a community library inside an office building:

- Employees (clients) can borrow books (files) or use the office printer.
- But the library is only for people inside the building (private network), not outsiders on the Internet.

5.2 APPLICATION LAYER PROTOCOLS

REMOTE DESKTOP PROTOCOL

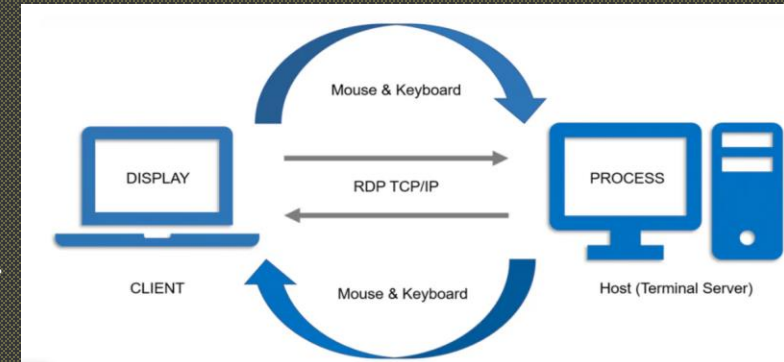
- RDP (Remote Desktop Protocol) lets you access and control a Windows computer remotely.
- You see the remote computer's graphical user interface (desktop) and can use it as if you were sitting in front of it.

Features

- Remote Access: Run Windows applications and access files from anywhere.
- Administration: Network administrators use RDP to manage servers and workstations remotely.
- Interactivity: Supports keyboard, mouse, and screen sharing.
- Port: Uses TCP port 3389.

Example

- A system admin works from home but needs to check logs on a server at the office.
 - They open the Remote Desktop Connection app on their PC.
 - Enter the server's IP address and login credentials.
 - The server's desktop appears on their screen → they can manage it as if physically present.



<https://ms.codes/blogs/remote-desktop/how-does-windows-remote-desktop-work>

RDP is like a remote control for your computer:

- You are holding the controller (client computer).
- The TV screen (remote computer's desktop) responds to every button press (keyboard/mouse).

5.2 APPLICATION LAYER PROTOCOLS

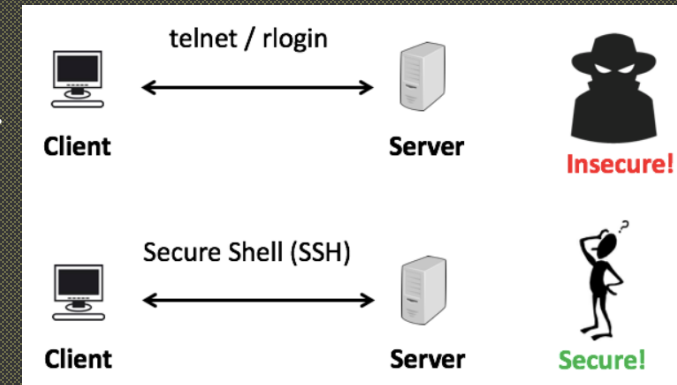
TELNET AND SECURE SHELL PROTOCOLS

Telnet

- Remote access to devices (e.g., routers, switches, servers) via a command-line interface.
- Port: Uses TCP port 23.
- Security: Not secure → data, including usernames and passwords, is sent in plain text.
- Use Case: Legacy systems or closed/private networks where security is less of a concern.
-

SSH (Secure Shell)

- Same as Telnet (remote device management via CLI) but with encryption.
- Port: Uses TCP port 22.
- Security: Provides a secure, encrypted channel for communication.
- Use Case: Preferred by network admins for securely managing servers, routers, and switches over both LAN and Internet.



<https://www.mysoftkey.com/security/ssh-secure-shell-protocol-for-secure-remote-login/>

- Telnet = Talking on a walkie-talkie in public → anyone nearby can overhear.
- SSH = Talking over a secure phone line → private and encrypted.
- CLI = Command Line Interface

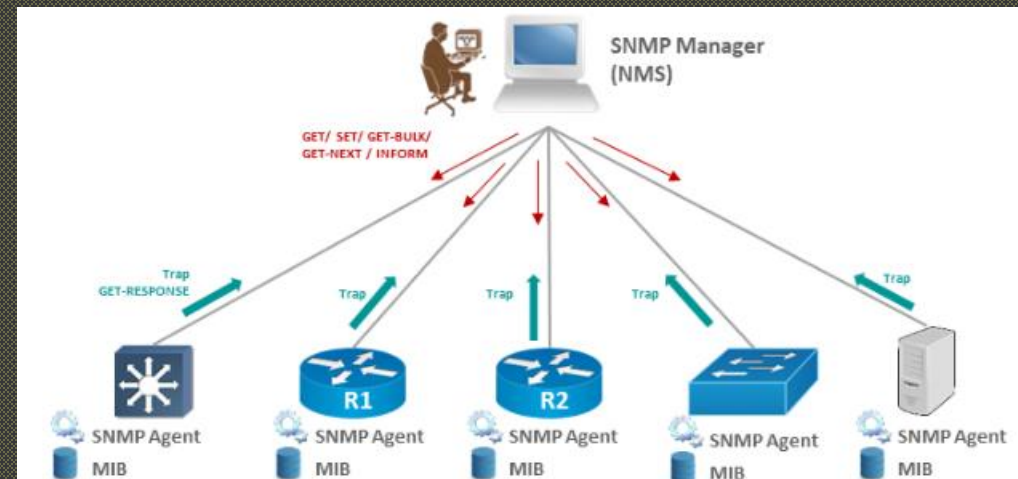
5.2 APPLICATION LAYER PROTOCOLS

SNMP PROTOCOLS

- Simple Network Management Protocol enables network administrators to monitor, manage, and control devices across a network.

Components

- Manager (NMS): A central application or tool (e.g., SolarWinds, Nagios, PRTG) that collects data, analyzes trends, and displays network health.
- Agent: A software process running on the managed device that maintains local data and communicates with the manager.
- Management Information Base (MIB): The structured database of managed objects, defining what information can be probed or modified on each device.



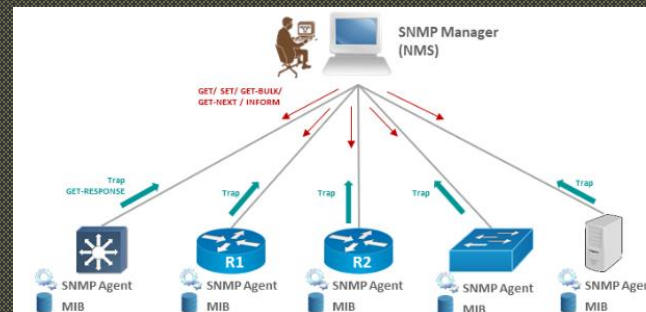
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5.2 APPLICATION LAYER PROTOCOLS

SNMP PROTOCOLS

How SNMP Works

- Manager - Agent Model
 - The manager (client) sends queries to the agent (server), and the agent responds with the requested information or performs actions.
- Management Information Base (MIB)
 - Every SNMP-enabled device has a MIB, a standardized database of variables (like CPU load, interface status, or error counts) that the manager can read or update.
- Communication Methods
 - Get Requests: The manager retrieves information from the agent's MIB.
 - Set Requests: The manager modifies a value in the MIB, effectively changing device behavior or configuration.
 - Traps/Notifications: Agents send unsolicited messages to the manager to report significant events (e.g., link failure, overheating) without waiting for a query.
- Transport Protocol
 - SNMP typically uses UDP (User Datagram Protocol) for communication, ensuring lightweight and efficient message delivery.



<https://networkwalks.com/snmp-simple-network-management-protocol/>

5.2 APPLICATION LAYER PROTOCOLS

SNMP PROTOCOL

- A protocol used to monitor, manage, and control network devices (e.g., routers, switches, printers, servers).
- Works in a client/server model:
 - Agent: Runs on the network device, collects information (e.g., CPU usage, interface status, errors).
 - Network Management Station (NMS): The central server/software that queries agents and stores/analyzes the data.

Ports

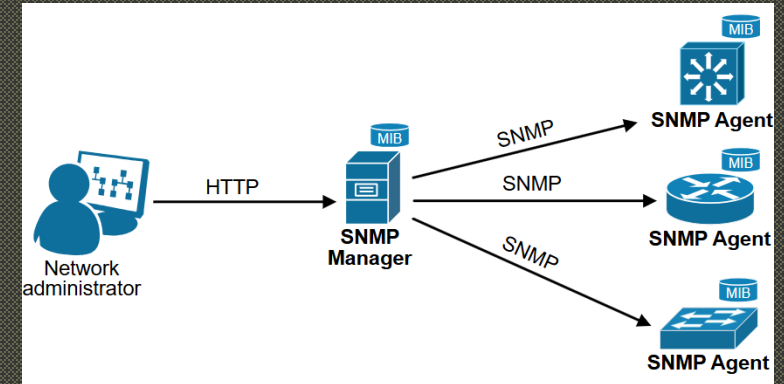
- UDP 161 → Used by the NMS to query agents.
- UDP 162 → Used by agents to send *traps* (alerts) back to the NMS.

Functions of SNMP

- Monitoring: Check device health (e.g., up/down status, bandwidth usage).
- Management: Change device configurations remotely.
- Alerts (Traps): Devices automatically notify the NMS when critical events occur (e.g., link failure).

SNMP (Simple Network Management Protocol) is like a security guard system:

- Each guard (agent) watches over a device.
- They report back to a central security office (NMS).
- If something urgent happens (a fire alarm), the guard immediately sends an alert (trap).



<https://www.networkacademy.io/ccna/network-services/simple-network-management-protocol-snmp>

- The SNMP manager = Network Management Station (NMS)
- The SNMP agent is a software module installed in the device
- The Management Information Base (MIB) is the database of the device's variables.

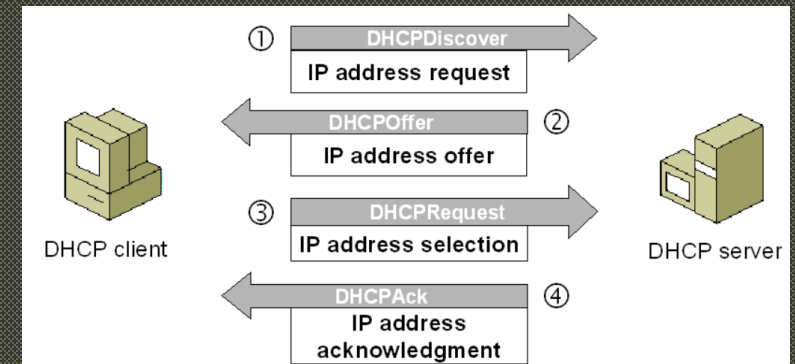
5.2 APPLICATION LAYER PROTOCOLS

DHCP PROTOCOL

- Dynamic Host Configuration Protocol automatically assigns IP addresses and network settings (like subnet mask, gateway, DNS servers) to devices on a network.
- Eliminates manual configuration of devices and keeps track of which IP addresses are assigned to which devices.
 - Example: In a school with 100 computers, manually setting IP addresses for each would be time-consuming. DHCP assigns them automatically when devices connect.

How DHCP Works

- Basic Process:
 - A device (client) sends a broadcast message to find a DHCP server when it connects to the network.
 - The DHCP server responds with an available IP address and other settings from its configured pool.
 - The client accepts the IP address and "leases" it for a set period. It is acknowledged by the server.
 - The client renews the lease periodically or requests a new IP if the lease expires.
- Example: When you connect your laptop to a Wi-Fi network at a café, it automatically gets an IP address from the café's DHCP server, allowing you to access the internet without manual setup.

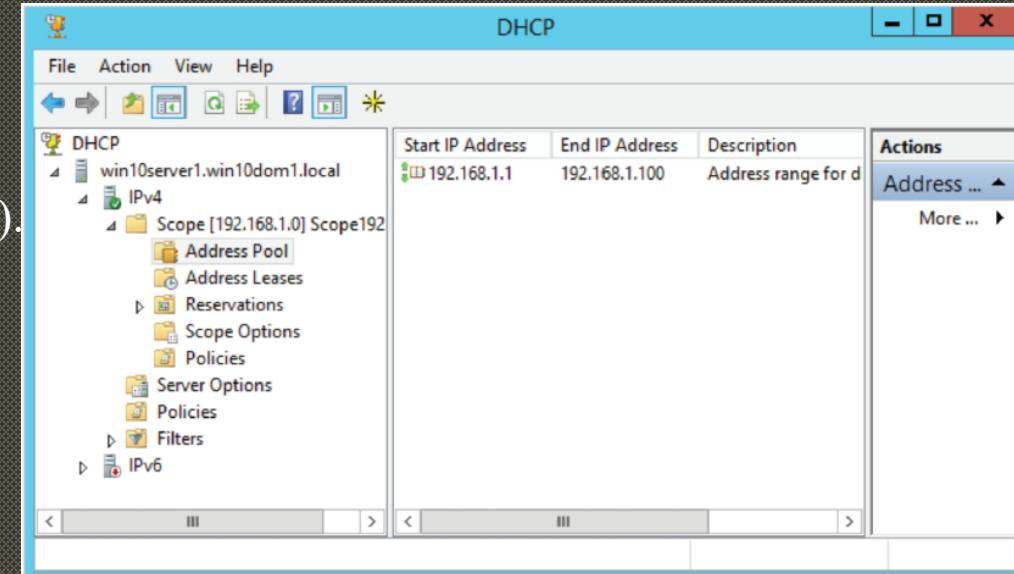


5.2 APPLICATION LAYER PROTOCOLS

DHCP PROTOCOL

Components of a DHCP Server

- IP Address Scope:
 - A range of IP addresses the server can assign (lease) to clients.
 - Includes:
 - Start and end IP addresses (e.g., 192.168.1.1 to 192.168.1.100).
 - Subnet mask (e.g., 255.255.255.0).
 - Lease duration (e.g., 1 day, 1 week, or unlimited).
 - Scope Options: Additional settings provided to clients, such as:
 - Default gateway (router's IP address, e.g., 192.168.1.1).
 - DNS servers (e.g., 8.8.8.8 for Google's DNS).
 - Domain name (e.g., example.com).
- Example: A DHCP server in an office might have a scope of 192.168.1.1 to 192.168.1.200, assigning each device an IP address, a subnet mask, and the address of the office's router.

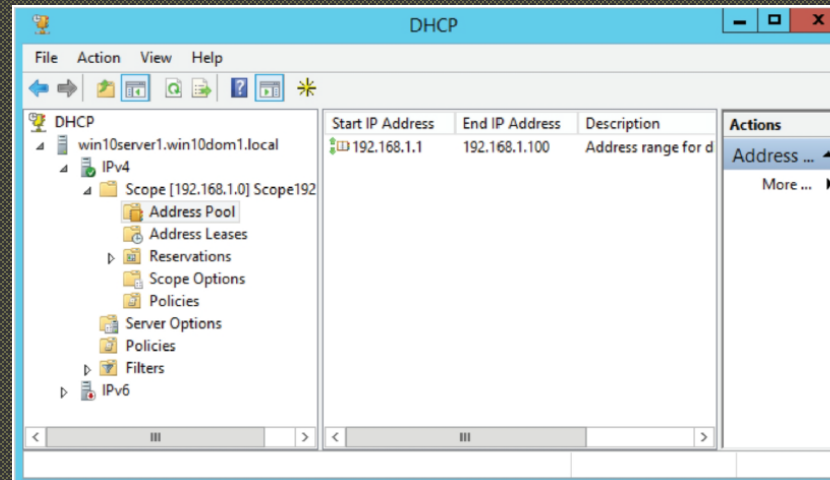


5.2 APPLICATION LAYER PROTOCOLS

DHCP PROTOCOL

Components of a DHCP Server

- Exclusions:
 - IP addresses within the scope that are not assigned by DHCP, reserved for devices with static (manually set) IPs.
 - Example: If a server in the network uses a fixed IP like 192.168.1.5, it is excluded from the DHCP scope to avoid conflicts.
- DHCP Server Service:
 - Runs in the background on a server, listening for client requests on UDP port 67.
 - Assigns IP addresses, tracks leases, and responds to renewal/release requests.
 - Stores lease details (IP, device name, MAC address, expiration time) in a database.
 - Example: A network admin can check the DHCP server's database to see that computer "PC-123" is using IP 192.168.1.15, leased until tomorrow.



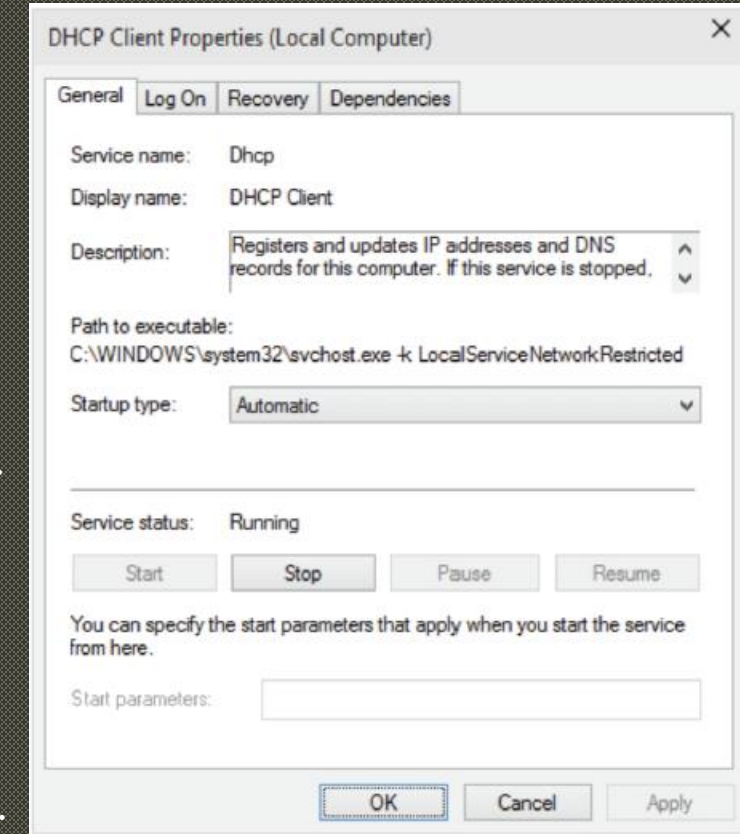
5.2

APPLICATION LAYER PROTOCOLS

DHCP PROTOCOL

DHCP Lease Process

- Lease Assignment:
 - A client broadcasts a request for an IP address.
 - The DHCP server assigns an available IP from its scope and sends it to the client.
- Lease Renewal:
 - At 50% of the lease duration (e.g., 12 hours on a 1-day lease), the client tries to renew the lease with the same server.
 - If no response, it tries again at 87.5% of the lease duration (e.g., 21 hours).
 - If the lease expires without renewal, the client requests a new IP address.
- Example: A laptop with a 1-day lease (24 hours) will try to renew its IP after 12 hours. If the DHCP server is down, it tries again at 21 hours. If it still fails, it requests a new IP.
- Automatic Private IP Addressing (APIPA):
 - If no DHCP server responds, a device may assign itself an IP from the 169.254.x.x range.
 - Used for small, non-internet-connected networks; not routable (can not access the internet).
 - Example: If your home Wi-Fi router's DHCP server fails, your computer might assign itself an IP like 169.254.1.1, but you will not be able to browse the internet.



5.2 APPLICATION LAYER PROTOCOLS

DHCP PROTOCOL

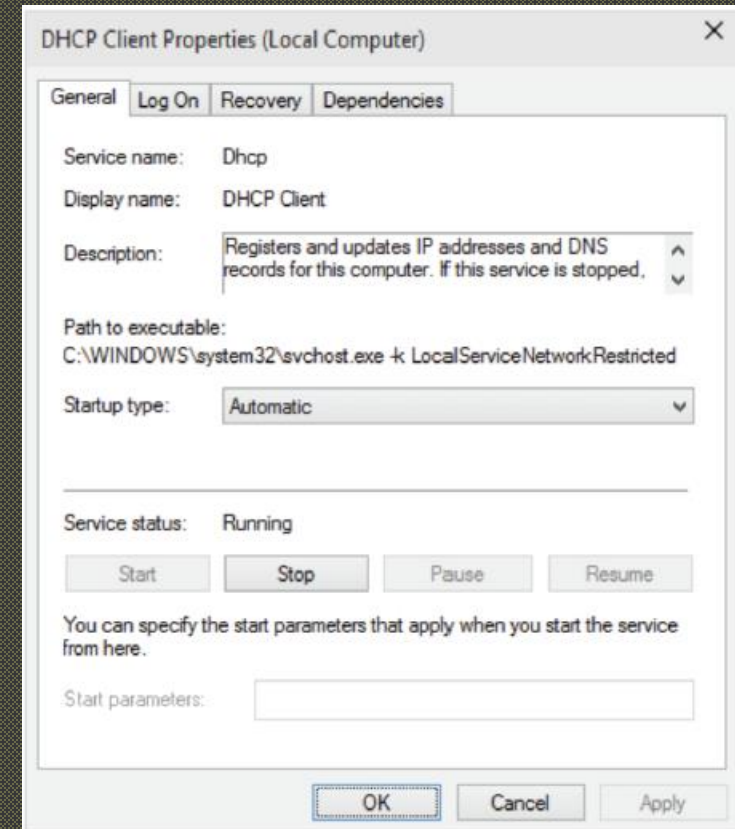
Technical Details

- Protocol Used: DHCP uses UDP (User Datagram Protocol) for communication because:
 - Messages are short, and UDP is fast and lightweight.
 - Clients and servers are typically on the same network.
- Port: DHCP server listens on UDP port 67; clients use UDP port 68.
- Supported Platforms: Most operating systems (Windows, Linux, macOS) and routers include DHCP client and server software.

Does DHCP help with saving IPv4 addresses?

DHCP is like a hotel receptionist:

- Guests (devices) check in and request a room (IP address).
- The receptionist (DHCP server) assigns one from the available rooms (scope).
- The room is theirs only for the length of their stay (lease time).
- VIP guests (reservations) always get the same room.



5.2 APPLICATION LAYER PROTOCOLS

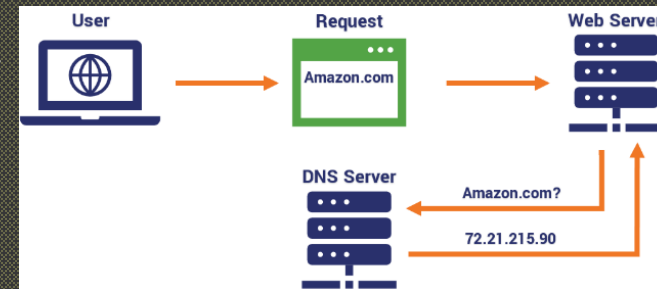
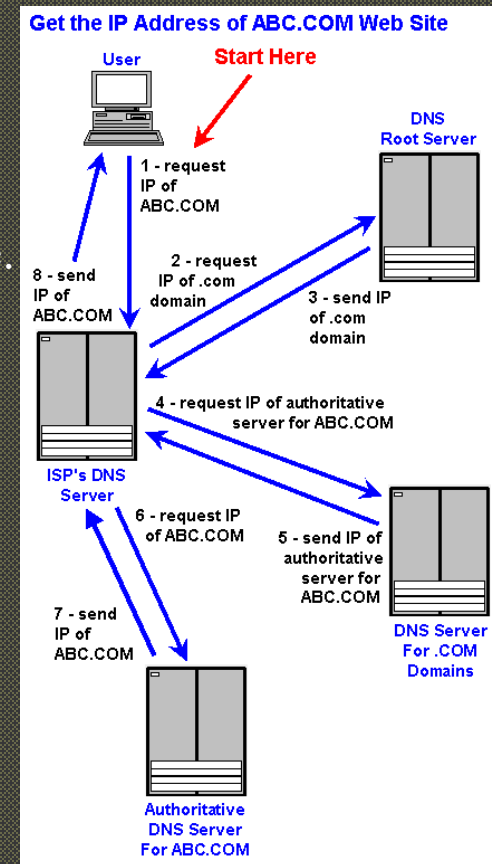
DNS PROTOCOL

The Domain Name System (DNS) is a protocol that translates user-friendly computer names (like www.google.com) into numerical IP addresses (like 142.250.190.78) that devices use to communicate over a network. It acts like a phonebook for the internet, making it easier to access websites and other network resources without memorizing complex IP addresses.

- Eliminates the need to remember numerical IP addresses, making network navigation easier.
- How it works:
 - You enter a domain name (e.g., www.ABC.com) in a browser.
 - The DNS client on your device contacts a DNS server to resolve the name to an IP address.
 - The DNS server responds with the IP address, allowing your browser to connect to the website.

Features of DNS

- Uses UDP Protocol:
 - DNS uses UDP (User Datagram Protocol) because messages are short (usually one packet), and UDP is fast and lightweight compared to TCP.
 - Example: A DNS query asking for the IP of www.google.com is a small, single message, so UDP is efficient.
- Port: DNS servers listen on UDP port 53 for client requests.

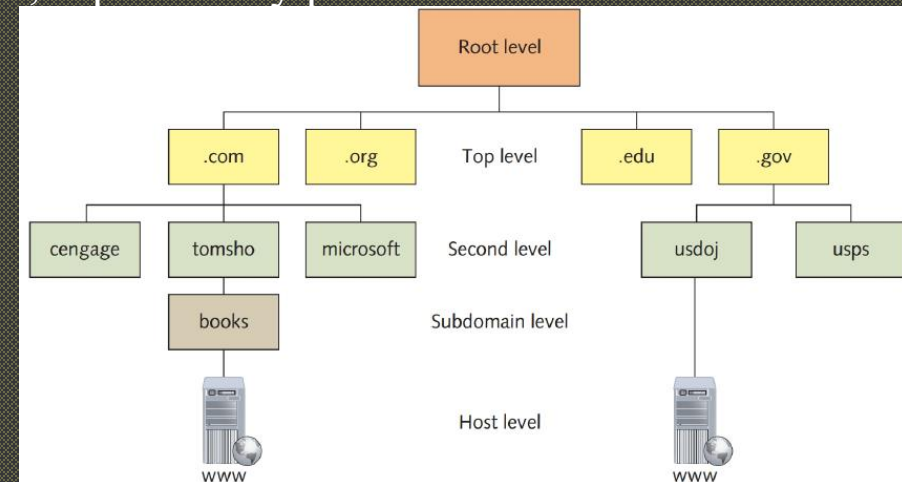


5.2 APPLICATION LAYER PROTOCOLS

DNS PROTOCOL

DNS Structure: Hierarchical Organization

- No single DNS server stores all internet addresses; each manages a specific domain or zone.
- Example: A DNS server for xyzcorp.com only stores addresses for that domain, not the entire internet.
- Tree-Like Hierarchy: DNS is organized like a tree with levels that form a domain name.
 - Levels:
 - Root: The top of the hierarchy (not visible in domain names).
 - Top-Level Domain (TLD): Categories like .com, .org, .edu, .gov, or country codes like .uk, .ca.
 - Second-Level Domain: Usually the name of a company or organization (e.g., google, amazon).
 - Subdomain (optional): A subdivision, like a department (e.g., books, mail).
 - Host: The specific device or service, like www or ftp.
 - Fully Qualified Domain Name (FQDN): Combines all levels, separated by periods.
 - Example: In www.books.tomsho.com:
 - .com = Top-level domain
 - tomsho = Second-level domain
 - books = Subdomain
 - www = Hostname



usdoj - U.S. Department of Justice
usps - U.S. Postal Service

5.2 APPLICATION LAYER PROTOCOLS

DNS PROTOCOL

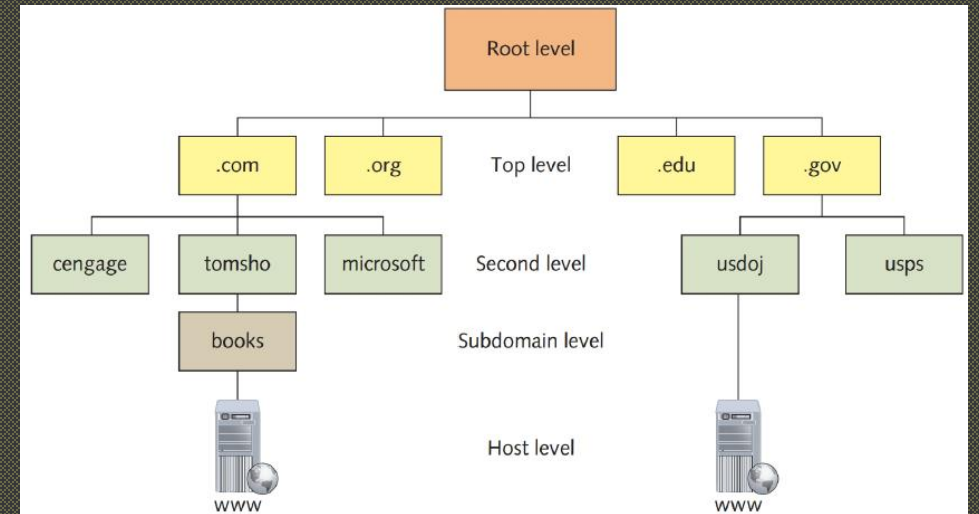
DNS Server and Zones

- DNS Server:
 - Maintains a database of domain names and their corresponding IP addresses for one or more zones.
 - A zone is a portion of the DNS namespace (e.g., xyzcorp.com).

- Example of Zones in Action

The company XYZ Corp with the domain xyzcorp.com.

- Zone 1: xyzcorp.com
 - Authoritative for the root of the company's namespace.
 - Contains records like:
 - www.xyzcorp.com → 192.168.10.10
 - mail.xyzcorp.com → 192.168.10.20
 - ftp.xyzcorp.com → 192.168.10.30
- Zone 2: books.xyzcorp.com
 - A subdomain of xyzcorp.com.
 - Could be hosted on a separate DNS server responsible only for the "books" division.
 - Contains records like:
 - store.books.xyzcorp.com → 192.168.20.5
 - api.books.xyzcorp.com → 192.168.20.6



5.2 APPLICATION LAYER PROTOCOLS

DNS CLIENT

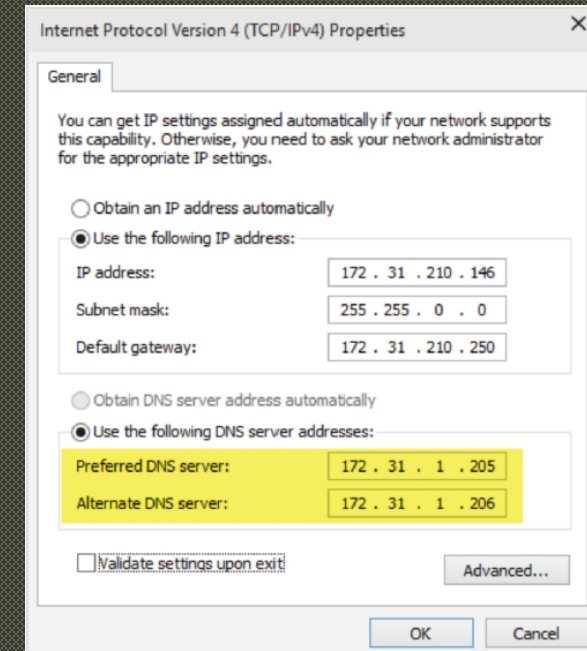
- A DNS Client (also called a resolver) is software on your computer that:
 - Talks to DNS servers.
 - Converts computer names → IP addresses.
- Runs as a service in Windows (can be seen/configured in *Services* control panel).

DNS Cache (Resolver Cache)

- Stores results of recent DNS lookups.
- Speeds up communication (no need to ask DNS server again if the name was resolved recently).
- Example:
 - You visit www.google.com once → IP saved in cache.
 - Next visit → computer uses cache instead of contacting DNS server again.

DNS Server Configuration on Client

- A computer must know at least one DNS server address to work.
- In Windows:
 - Preferred DNS server = main server used first.
 - Alternate DNS server = backup server if preferred is unavailable.



5.2 APPLICATION LAYER PROTOCOLS

DNS CLIENT

Why DNS needs suffixes

- Computers find each other using FQDNs (Fully Qualified Domain Names) like server1.company.com.
- But users are lazy - they often just type server1.
- To make this work, Windows (or any OS) automatically adds a DNS suffix to complete the name.

Primary DNS Suffix

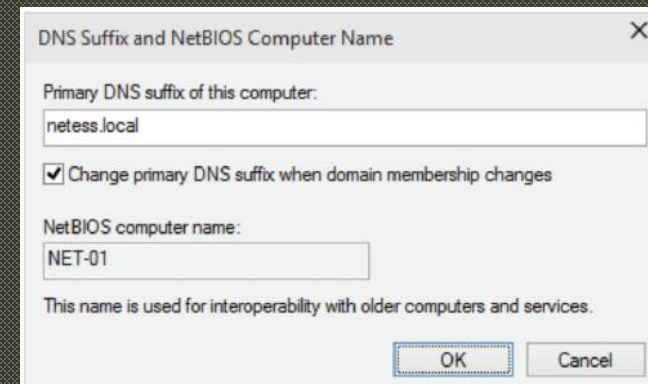
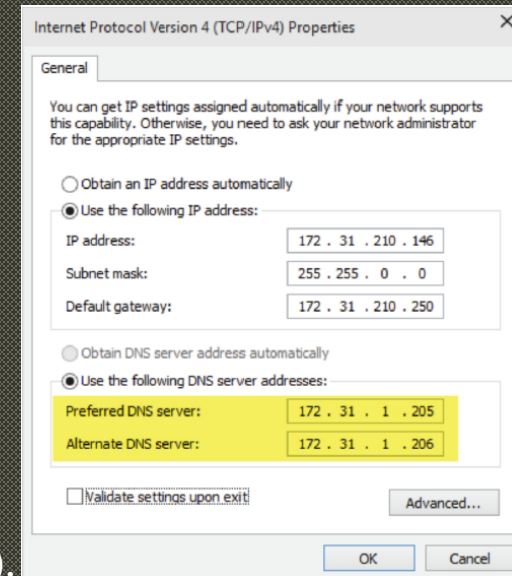
- This is the default suffix that gets added.
- Usually set when the computer joins a domain (e.g., netess.local).
- So:
 - You type: server
 - Resolver changes it to: server.netess.local

Multiple DNS Suffixes

- In bigger organizations, there may be several domains (like sales.company.com, hr.company.com).
- The resolver can try them one by one until it finds a match.

Example:

- You type: server1
- Resolver tries in order:
 - server1.sales.company.com
 - If not found → server1.hr.company.com



5.2 APPLICATION LAYER PROTOCOLS

DNS CLIENT

- Devices often get new IP addresses when they move around or reconnect (especially with DHCP).
- Without updates, DNS would point to the wrong IP.

What DDNS Does

- Dynamic DNS (DDNS) lets devices automatically tell the DNS server:
- “Hello, my IP has changed - update your records.”
- This keeps the DNS database correct without needing an admin to fix it manually.

When It Happens

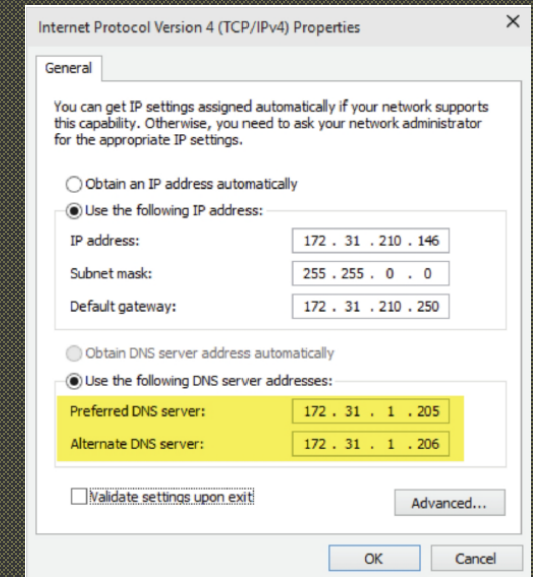
- IP address changes (new DHCP lease).
- Hostname changes (device renamed).

Example

- Your laptop connects at home: gets IP 192.168.1.20.
- Later, you take it to the office: new IP 10.0.0.55.
- Laptop uses DDNS to notify the DNS server → DNS updates to show the laptop's new IP.
- Now other computers can still find it by name (not the old, wrong IP).

Think of the DNS client as your personal assistant:

- You ask it for someone's phone number (IP).
- If it remembers (cache), it answers immediately.
- If not, it looks in the company phonebook (DNS server).
- If the name is incomplete, it guesses the “last name” (suffix).
- If you change your number, the assistant updates the phonebook automatically (DDNS).



5.2 APPLICATION LAYER PROTOCOLS

DNS SERVER

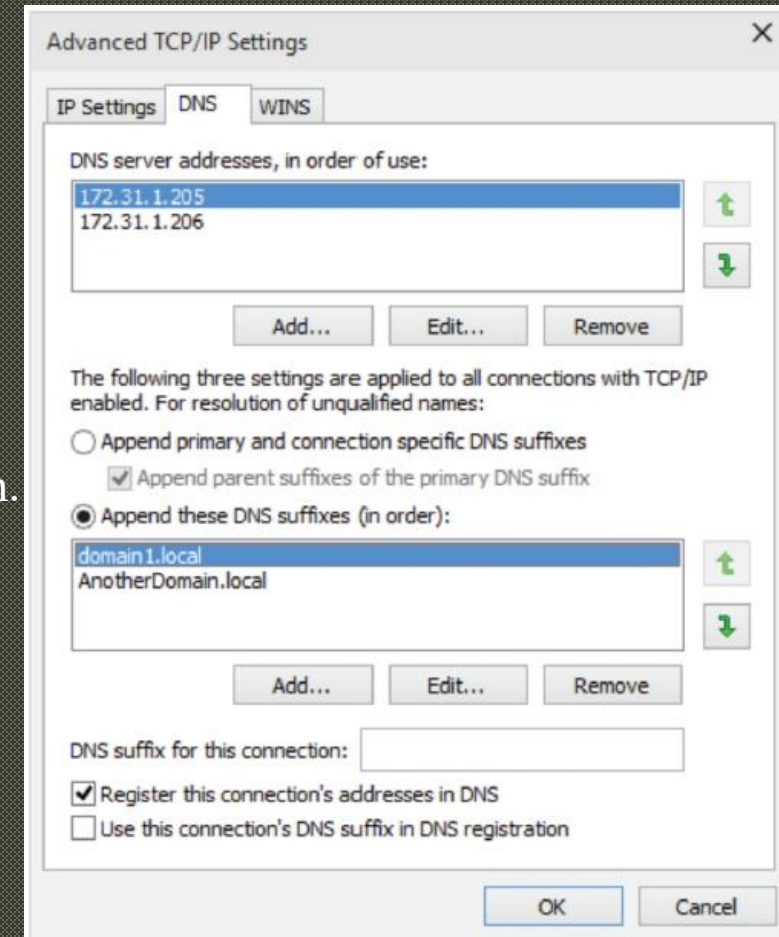
Role of DNS Server

- A DNS server translates computer/domain names → IP addresses (and vice versa).
- Used for:
 - Internet name resolution (e.g., `www.google.com`).
 - Local resource resolution (e.g., `printer1.office.local`).

Main Components of a DNS Server

a. DNS Zones

- A zone is like a database that stores hostname-IP address pairs for a specific domain.
- Example: A `company.com` DNS zone might contain:
 - `www.company.com` → `203.0.113.10`
 - `mail.company.com` → `203.0.113.20`
- Zones can be:
 - Created manually by an administrator.
 - Updated automatically by devices using Dynamic DNS (DDNS).
- If a client asks for `mail.company.com`:
 - DNS server checks its zone records.
 - If found → responds with the correct IP.



5.2

APPLICATION LAYER PROTOCOLS

DNS SERVER

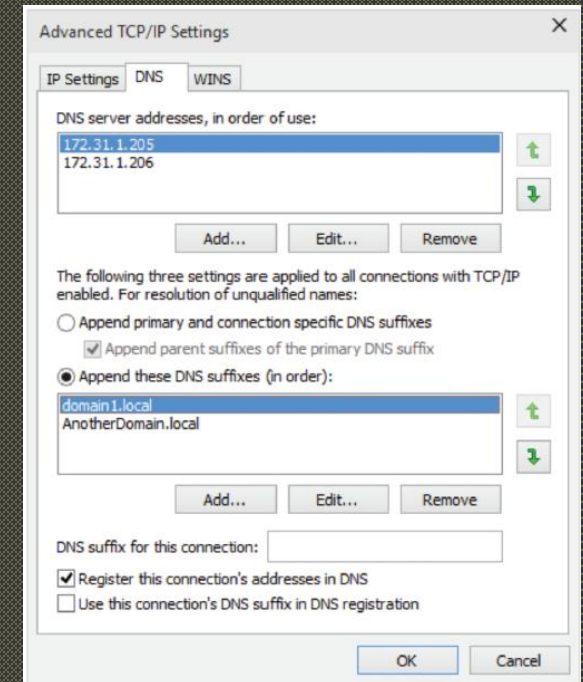
Main Components of a DNS Server

b. Resource Records

- Resource Records (RRs) = entries inside zones.
- Types of records (examples):
 - A record → Maps a hostname to an IPv4 address.
 - Example: `www.company.com` → `192.168.1.10`
 - AAAA record → Maps a hostname to an IPv6 address.
 - Example: `www.company.com` → `2001:db8::1`
 - MX record → Points to the mail server that handles email.
 - Example: `company.com` → `mail.company.com`
 - NS record → Says which name servers are in charge of the domain.
 - Example: `company.com` → `ns1.company.com`

c. Cache

- DNS servers (and even your PC) keep a temporary copy of past DNS query results.
Prevents contacting other servers again for the same query.
- Example:
 - First request:
 - User asks for `www.google.com`.
 - DNS doesn't know → asks another DNS server.
 - Result (e.g., IP = `142.250.190.4`) is stored in cache.
 - Next request (same name):
 - DNS finds it in cache.
 - Replies instantly without contacting external servers.
- Cached results expire after a set time (TTL → time-to-live).



5.2 APPLICATION LAYER PROTOCOLS

DNS SERVER

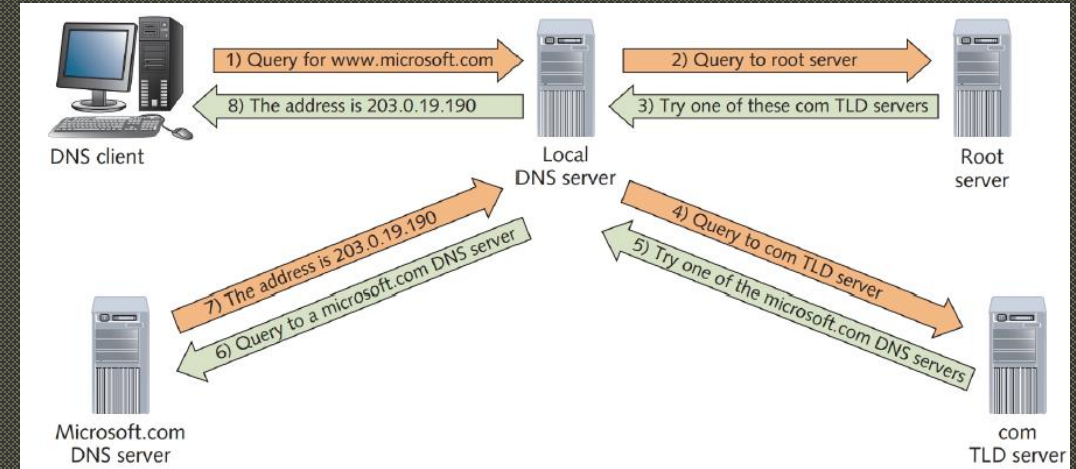
Main Components of a DNS Server

d. Root Hints

- A built-in file containing IP addresses of Internet root DNS servers.
- Used when:
 - DNS server has no local zone record.
 - No cache entry is available.
- Query path example:
 - User types `www.amazon.com`.
 - Local DNS server doesn't know → checks root servers.
 - Root server → tells which TLD server (.com) to contact.
 - TLD server → points to Amazon's DNS servers.
 - Amazon's server → responds with IP for `www.amazon.com`.

e. DNS Server Service

- Background service that:
 - Listens for DNS queries on UDP port 53.
 - Responds with the correct IP address (if it knows it).



5.2 APPLICATION LAYER PROTOCOLS

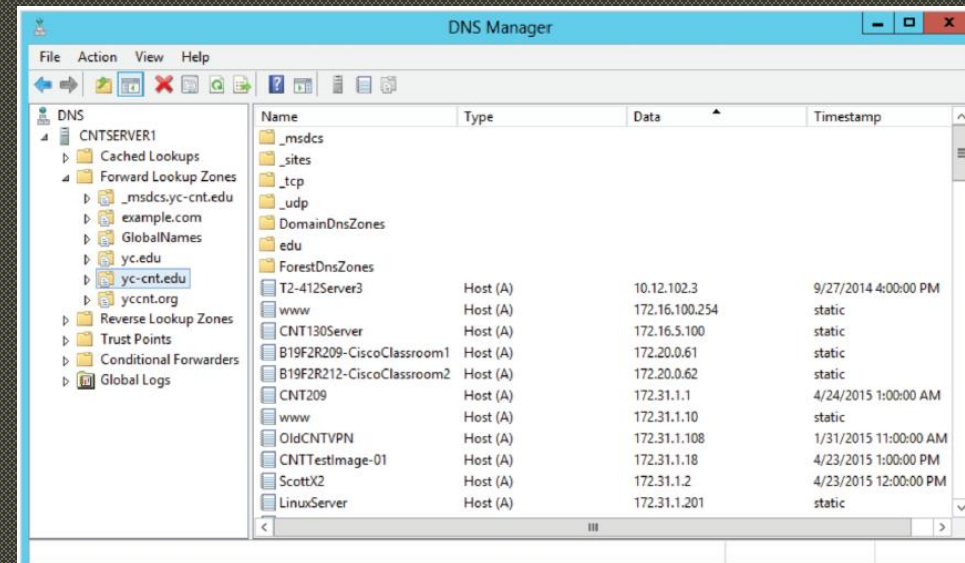
DNS SERVER

Types of DNS Servers

- Authoritative Server:
 - Stores original zone data (e.g., your company's DNS server for company.com).
- Caching-Only Server:
 - Doesn't have its own zones.
 - Just stores answers from other DNS servers.
 - Common for home/small business networks.
 - Benefits:
 - Reduces traffic to ISP DNS servers.
 - Answers faster for frequently visited sites.

Think of the DNS server as a phonebook office:

- Zones = local phonebook section (your city).
- Resource records = actual names and phone numbers.
- Cache = sticky notes of recently looked-up numbers.
- Root hints = master list of where to find other phonebooks worldwide.
- Caching-only server = a clerk who doesn't have their own book but remembers numbers once they've looked them up.



5.3 TRANSPORT LAYER PROTOCOLS

PORT NUMBERS

- A port number is like a door into a specific service or application on a device.
- An IP address = the building's street address (identifies the device on the network).
- A port number = the apartment number (directs data to the correct software/application inside the device).

Example:

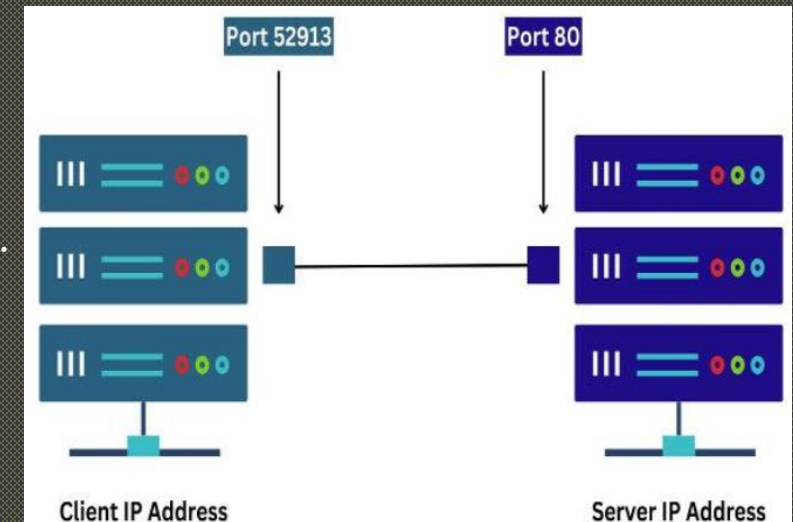
- Imagine sending a letter to an office building:
 - Street address (IP address): Gets the mail to the correct building.
 - Office number (Port number): Ensures the letter reaches the right office/person inside.

Why Do We Need Ports?

- A computer can run many applications at once (browser, email, online game, etc.).
- Ports make sure network traffic does not get mixed up.
- The operating system manages ports and decides which app should get which data.

Example:

- You open:
 - Web browser → uses port 80 (HTTP) or 443 (HTTPS).
 - Email client → uses port 25, 110, or 143.
 - Online game → might use port 3074.
- Thanks to ports, each program gets only its own data.



<https://www.geeksforgeeks.org/computer-networks/what-is-network-port/>

5.3 TRANSPORT LAYER PROTOCOLS

PORT NUMBERS

How Do Ports Work in Networking?

- Ports are included in the header of data packets.
- Transport protocols like TCP and UDP use them to deliver data correctly.

Example:

- When you visit `www.google.com`:
 - Your browser connects to IP address of Google's server.
 - It sends the request to port 443 (HTTPS) on that server.
 - The reply is sent back to your computer's temporary port number, so your browser gets it and not your email app.

Port Numbers and Security

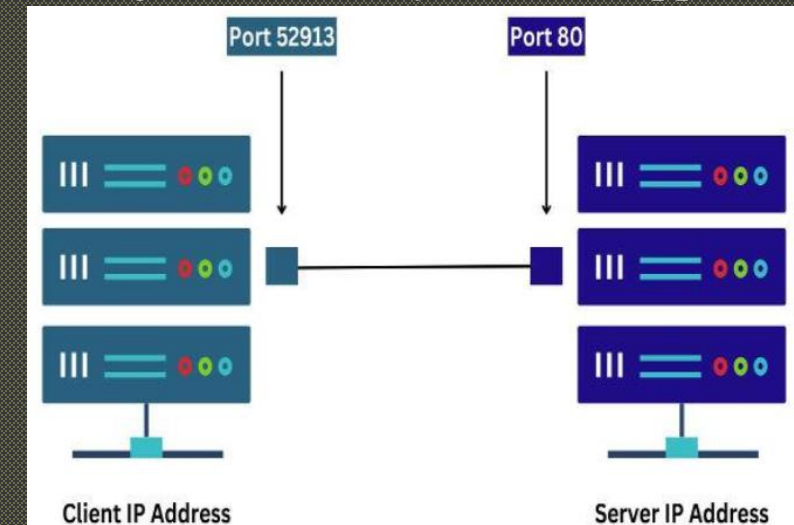
- Firewalls use port numbers to allow or block traffic.
- Attackers may scan ports to find open services they can exploit.
- Good practice:
 - Close unnecessary ports.
 - Only allow ports needed for your services.

Example:

- A web server only needs ports 80 (HTTP) and 443 (HTTPS) open.
- Closing unused ports (like 21 for FTP, if not used) reduces risks.

Common Port Numbers to Know:

- 20, 21 → FTP (File Transfer Protocol)
- 22 → SSH (Secure Shell)
- 25 → SMTP (sending emails)
- 53 → DNS (Domain Name System)
- 80 → HTTP (websites)
- 110 → POP3 (receiving emails)
- 143 → IMAP (receiving emails)
- 443 → HTTPS (secure websites)
- 3389 → RDP (Remote Desktop Protocol)



<https://www.geeksforgeeks.org/computer-networks/what-is-network-port/>

5.3 TRANSPORT LAYER PROTOCOLS

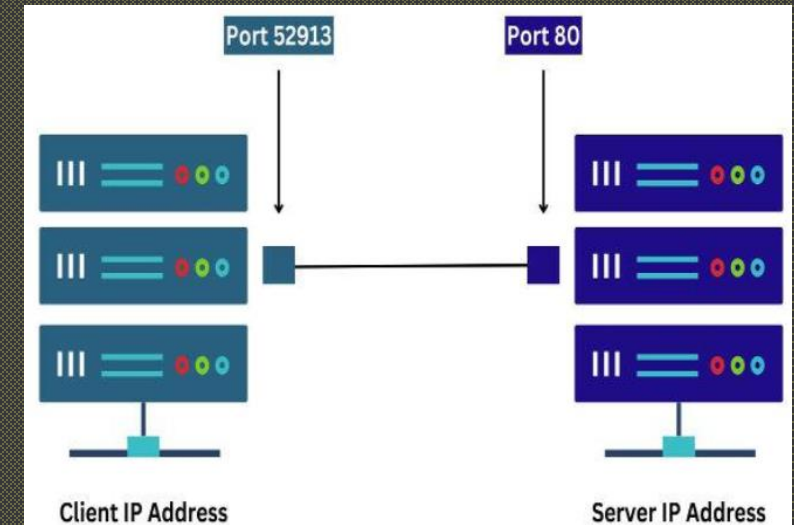
PORT NUMBERS

Port Number Ranges

- Port numbers are 16-bit numbers → range: 0 to 65,535.

They are divided into categories by IANA (Internet Assigned Numbers Authority):

- Well-Known Ports (0 - 1023)
 - Reserved for common services.
 - Often require admin/root permission to use.
 - Example:
 - 80 → HTTP (websites)
 - 443 → HTTPS (secure websites)
- Registered Ports (1024 - 49151)
 - Used by specific applications.
 - Officially registered but not as restricted.
 - Example:
 - 3306 → MySQL database
 - 3389 → RDP (Remote Desktop Protocol)
- Dynamic/Private Ports (49152 - 65535)
 - Used for temporary client-side connections.
 - Assigned automatically by the OS.
 - Example:
 - When requesting a webpage, a computer might use port 49500 to talk to the server's port 443.



<https://www.geeksforgeeks.org/computer-networks/what-is-network-port/>

5.3 TRANSPORT LAYER PROTOCOLS

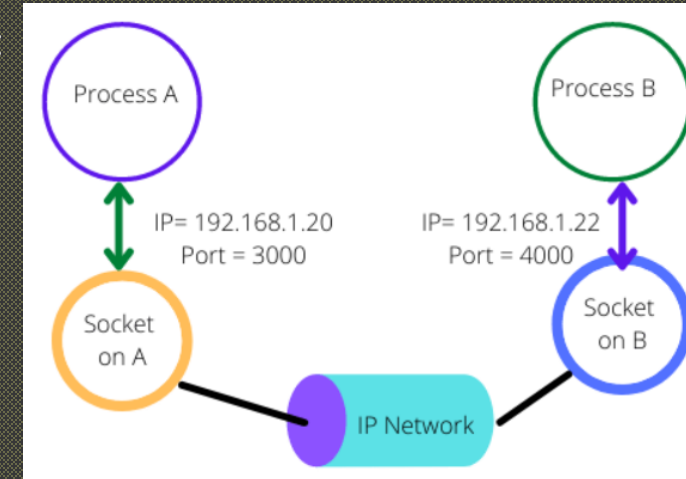
SOCKETS

A socket is the software endpoint of a two-way communication channel between two programs.

- A socket is uniquely defined by a pair: (IP address, Port number)
- Example: 192.168.1.20:80 represents a socket bound to IP address 192.168.1.20 on port 80.

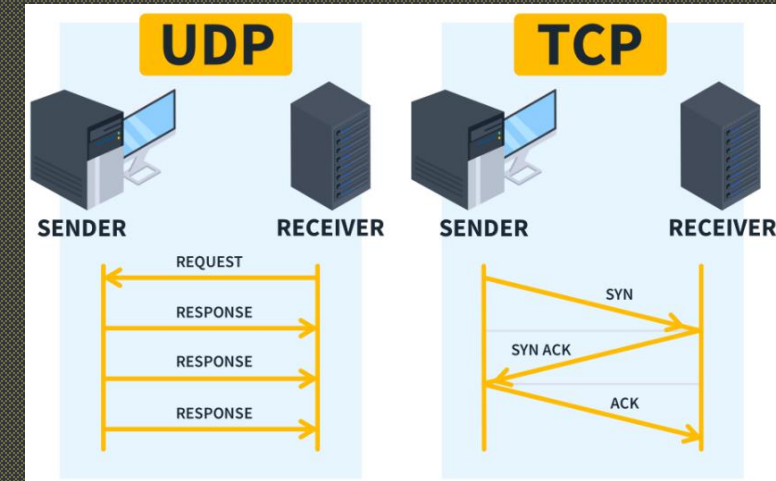
Example: Simple Client - Server Communication through an HTTP request (web browsing):

- Server Setup:
 - A web server (like Apache) creates a socket.
 - It binds port 80 on IP 192.168.1.10.
 - The server listens for incoming connections.
- Client Request:
 - Your browser (client) creates a socket.
 - It asks to connect to 192.168.1.10:80.
 - The OS establishes a TCP connection between the client socket and the server socket.
- Data Exchange:
 - Browser sends: GET /index.html HTTP/1.1.
 - Server reads this request through its socket, then responds with HTML content.
 - Data flows back and forth as long as the connection stays open.
- Closure:
 - When the browser finishes loading the page, it closes its socket.
 - The server may close its side, freeing resources.



5.3 TRANSPORT LAYER PROTOCOLS

- The Internet is unreliable: data can get lost, delayed, or arrive out of order.
- The Transport Layer ensures that data is delivered reliably and to the correct application.
- Without it, large networks like the Internet wouldn't work properly.
- Example: Imagine mailing a large book in pieces.
- The Network layer (IP) delivers each piece to the correct address.
- The Transport layer makes sure:
 - All pieces arrive.
 - They are put back in the correct order.
 - Nothing is missing.



<https://www.cbttuggets.com/blog/technology/networking/what-are-transport-layer-protocols>

- The Two Transport Layer Protocols
 - TCP (Transmission Control Protocol): Connection-oriented (like making a phone call).
 - Provides reliability → ensures data is received, correct, and in order.
 - Used for large or important data transfers.
 - Examples: Web browsing (HTTP/HTTPS), email (SMTP, IMAP, POP3), file transfer (FTP).
 - UDP (User Datagram Protocol): Connectionless (like sending a postcard).
 - Fast but not guaranteed → no check if data arrived.
 - Used for small, quick, real-time communication.
 - Examples: Online games, video calls, live streaming, DNS lookups.

5.3 TRANSPORT LAYER PROTOCOLS

UDP PROTOCOL

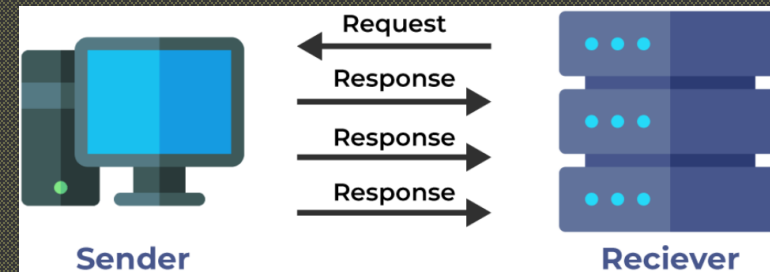
- User Datagram Protocol (UDP) is a simple, fast transport layer protocol.
- It is like sending a postcard: you send it without setting up a connection, and you have no guarantee it will arrive.

Characteristics & Why It's "Unreliable"

- Connectionless: No connection is established before sending data. Each packet (datagram) is independent.
- No Guaranteed Delivery: Packets can be lost, arrive out of order, or be duplicated, and UDP will not fix it.
- No Flow Control: It can send data faster than the receiver can handle, potentially overwhelming it.
- No Error Correction: If a packet is corrupted, UDP simply discards it. It does not ask for a retransmission.

Advantages & Why We Use It

- Low Overhead & Fast: The header is only 8 bytes, making it very small and efficient.
- No Connection Setup Delay: It can start sending data immediately.
- Good for Short Messages: Perfect for quick, one-off requests and responses.
- More Control for Applications: The application itself can handle any necessary error-checking in its own way.



5.3 TRANSPORT LAYER PROTOCOLS

UDP PROTOCOL

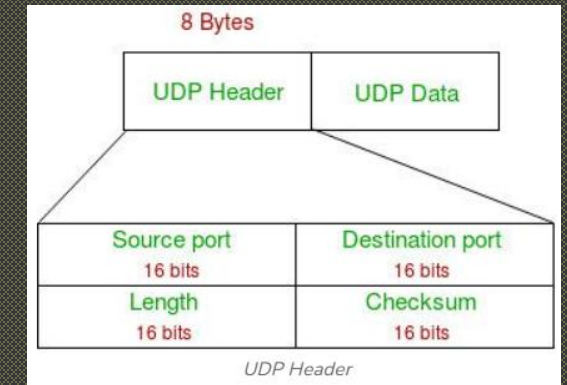
- The UDP Header (8 Bytes / 64 Bits)
- The header is very simple and has only four fields:
 - Source Port Number: Where the message came from.
 - Destination Port Number: Where the message is going.
 - Total Length: The total size of the UDP packet (header + data).
 - Checksum: (Optional) Used to check for errors in the header and data.

When to Use UDP (with Examples)

- DNS (Domain Name System) Lookups:
 - *Example:* Your computer asks a DNS server "What is the IP for google.com?" The request and response are single, short packets. Speed is more important than perfect reliability here.
- Real-Time Audio/Video Streaming:
 - *Example:* Services like Skype, Zoom, or online gaming. If a video frame is lost, it is better to skip it and keep the stream going smoothly rather than pausing to re-send it, which would cause a frozen screen.
- Simple Request/Response Protocols:
 - *Example:* DHCP (to get an IP address) and SNMP (for network management). These involve short, simple messages.
- Broadcasting/Multicasting:
 - *Example:* Sending data to multiple computers at once (e.g., live stock tickers). UDP natively supports this, while TCP does not.

Example: UDP header in hexadecimal format: CB84000D001C001

- The source port number: $CB84_{16} = 52,100_{10}$.
- The destination port number: $000D_{16} = 13_{10}$.
- The third four hexadecimal digits ($001C_{16}$) define the length of the - 28 bytes.
- The length of the data is the length of the whole packet minus the length of the header, or $28 - 8 = 20$ bytes.



<https://www.geeksforgeeks.org/computer-networks/user-datagram-protocol-udp/>

When NOT to Use UDP

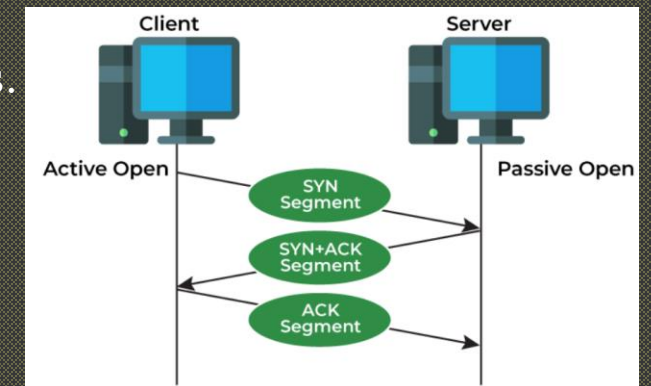
- Sending Large Files or Important Data:
 - *Example:* Downloading a file, loading a web page, or sending an email (SMTP). For these, you need the reliability of TCP to ensure every piece of data arrives correctly and in order. A corrupt or missing piece of a file would ruin it.
- When Data Order is Critical:
 - *Example:* If you are transferring a long text document, you need the paragraphs to arrive in the correct sequence.

5.3 TRANSPORT LAYER PROTOCOLS

TCP PROTOCOL

Characteristics & Why It is "Reliable"

- **Connection-Oriented:** A stable connection must be established before data is sent, and it is properly closed afterward.
 - **Example:** Like a phone call. You dial (request a connection), the other person says "Hello" (accepts the connection), you talk (transfer data), and then you both say "Goodbye" to end the call.
- **Reliable Delivery:** It guarantees that data sent is delivered accurately and in order. It uses acknowledgments (ACKs) and retransmissions.
 - **Example:** Like a signed delivery for an important package. You get a receipt (ACK) confirming it arrived.
- **Flow Control:** Prevents a fast sender from overwhelming a slow receiver. The receiver tells the sender how much data it can handle at a time (the "window").
- **Congestion Control:** Prevents a sender from overwhelming the network itself. It slows down the data rate if network traffic is heavy.
 - **Example:** Like traffic on a highway. If there is a jam (congestion), cars slow down and enter the highway more gradually to avoid making the jam worse. When the road clears, they can speed up again.
- **Stream-Oriented:** Data is treated as a continuous stream of bytes, not individual messages. TCP chops this stream into segments for transmission.
 - **Example:** Like water flowing from a tap into a hose. You don't think of the water as individual droplets, but as a continuous stream.



5.3 TRANSPORT LAYER PROTOCOLS

TCP PROTOCOL

Sockets and Connections

- For TCP to transfer data, both the sender and receiver must create a socket (an endpoint of communication). Multiple sockets can connect through the same port, which allows a server to handle many simultaneous TCP connections.

Tracking Data with Sequence Numbers

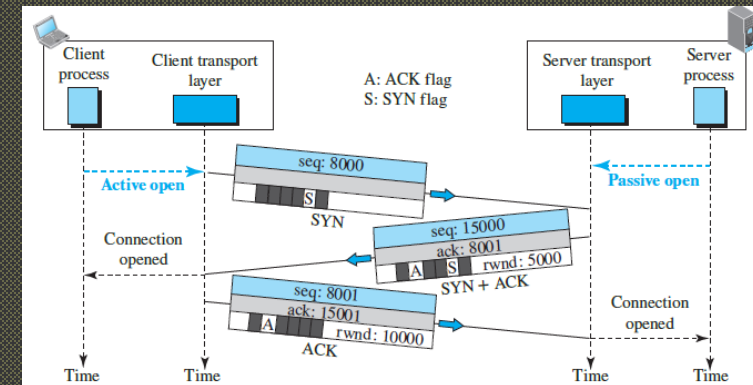
- TCP assigns a unique 32-bit sequence number to every byte of data in a connection.
- These sequence numbers ensure data is delivered in order and that no bytes are lost.
- The sequence number of the first byte in each segment is included in the TCP header.
- The receiver uses this value and increments it as each subsequent byte arrives.

Connection Setup:

- Before data transfer begins, the sender and receiver agree on their Initial Sequence Numbers (ISNs) during the three-way handshake (SYN → SYN-ACK → ACK). This process establishes a reliable, synchronized connection.

The "Three-Way Handshake" (Connection Establishment)

- SYN:** The client sends a "Synchronize Sequence Number" packet to the server. ("I want to talk.")
- SYN-ACK:** The server acknowledges the request and sends its own SYN. ("Okay, I'm ready to talk.")
- ACK:** The client acknowledges the server's SYN. ("Great, let's talk.")
 - Now the connection is established, and data transfer can begin.



5.3 TRANSPORT LAYER PROTOCOLS

TCP PROTOCOL

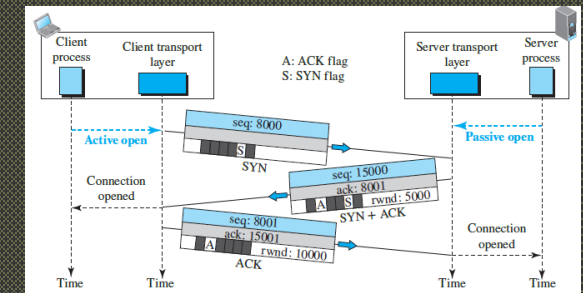
Acknowledgments (ACKs)

- When the receiver successfully gets data, it responds with an acknowledgment number (ACK).
- The ACK represents the next byte expected, which is equal to the last byte received + 1.

Example:

- Suppose 10 bytes are sent, starting with sequence number 33.
- Bytes are numbered from 33 to 42.
- If all 10 bytes arrive correctly, the receiver sends back ACK = 43.
- This tells the sender: *“I’ve received everything up to 42; please send me byte 43 next.”*

If a segment is missing or corrupted, the ACK points to the first missing byte, signaling the sender to retransmit from that point.



Sequence Numbers vs. Acknowledgment Numbers

- Sequence Number → Identifies bytes being sent (outgoing data).
- Acknowledgment Number → Identifies bytes successfully received (incoming data).
- These two values are independent, since data flows in both directions.

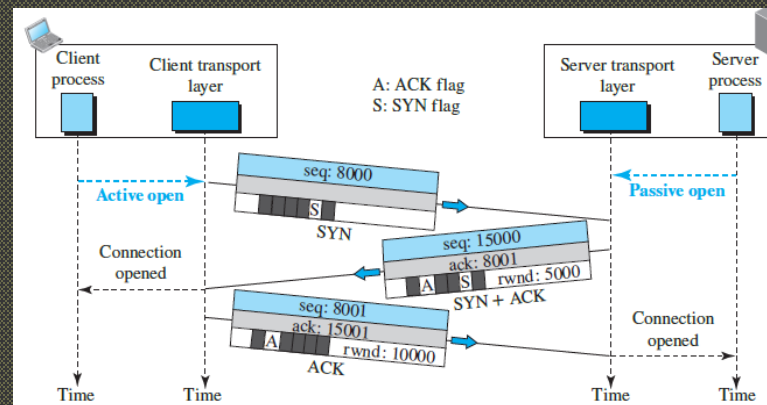
5.3

TRANSPORT LAYER PROTOCOLS

TCP PROTOCOL

When to Use TCP

- Web Browsing (HTTP/HTTPS):
 - *Example:* Loading a web page. You need every part of the page (text, images, code) to load completely and in the correct order. A missing image or scrambled text would ruin the experience.
- File Transfers (FTP, File Downloads):
 - *Example:* Downloading a software update or a movie file. You need every byte of the file to be correct.
- Email (SMTP):
 - *Example:* Sending an important document as an email attachment. You need reliability to ensure the entire document is sent correctly.
- Remote Access (SSH):
 - *Example:* A system administrator logging into a server remotely. Every keystroke and command must be transmitted reliably and in order.



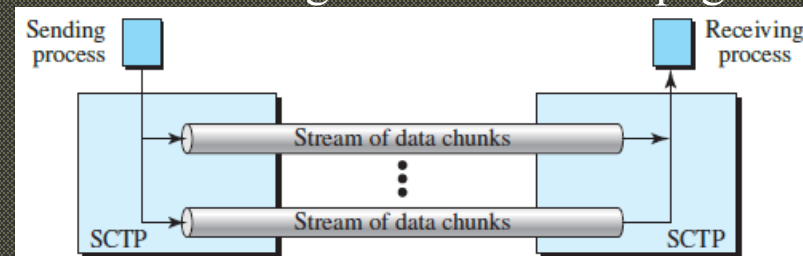
5.3 TRANSPORT LAYER PROTOCOLS

SCTP PROTOCOL

- Stream Control Transmission Protocol (SCTP) is a new TCP/IP protocol that combines the best features of TCP and UDP.
- It is especially good for real-time, multimedia communication that needs to be reliable but also fast.

Features & How It Improves on TCP/UDP

- Connection-Oriented & Reliable (like TCP):
 - It establishes a connection (called an "association") and guarantees that data arrives correctly and in order, using acknowledgments and retransmissions.
- Message-Oriented (like UDP):
 - It preserves message boundaries. If you send three distinct messages, the receiver gets three distinct messages. TCP would treat them as a continuous stream of bytes.
- Multi-Streaming (The "Head-of-Line Blocking" Solver):
 - A single SCTP connection can have multiple independent streams of data within it.
 - Example: Imagine loading a web page with text and images. In TCP, if one image packet is lost and being re-sent, it can block the text and all other images from being delivered, freezing the page load. In SCTP, the text, image 1, and image 2 could each be in their own stream. A loss in the "image 1" stream wouldn't block the delivery of the "text" stream or the "image 2" stream. The page would still load and be usable.



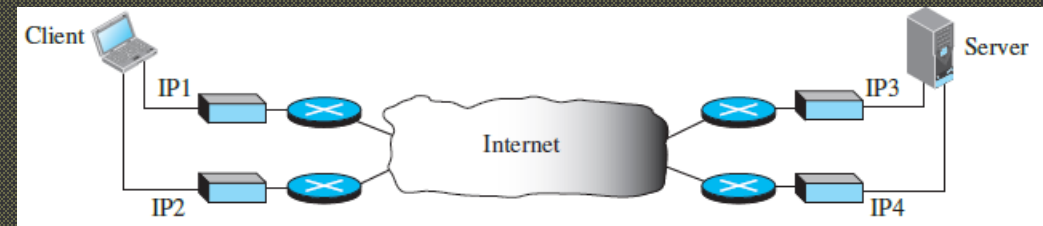
5.3 TRANSPORT LAYER PROTOCOLS

SCTP PROTOCOL

Features & How It Improves on TCP/UDP

- Multihoming (Fault Tolerance):
 - A single SCTP connection can have multiple network paths. For example, your phone could be connected to both Wi-Fi and a cellular network simultaneously for the same call.
 - Example: A VoIP call (like WhatsApp or Skype call). If your Wi-Fi signal becomes weak or drops, SCTP can almost instantly switch the call to your cellular data connection without dropping the call. This provides much better resilience.
- Four-Way Handshake (More Secure Connection Setup):
 - In SCTP's four-way handshake, the "cookie" is included as a temporary ticket given to a new connection request.
 - The server does not immediately reserve resources (like memory) for every request - instead, it sends the client a cookie.
 - The client must send this cookie back to prove it is a real, valid request before the server commits resources.

The SCTP cookie is a safety check during connection setup. It makes sure only real clients get connected, protecting the server from being flooded with fake requests.
- Selective Acknowledgments (SACK):
 - The receiver can tell the sender exactly which pieces of data are missing, not just the next expected one. This allows for more efficient retransmission of only the lost data.



5.3 TRANSPORT LAYER PROTOCOLS

SCTP PROTOCOL

When is SCTP Used? (Real-Life Applications)

SS7 – Signal System 7

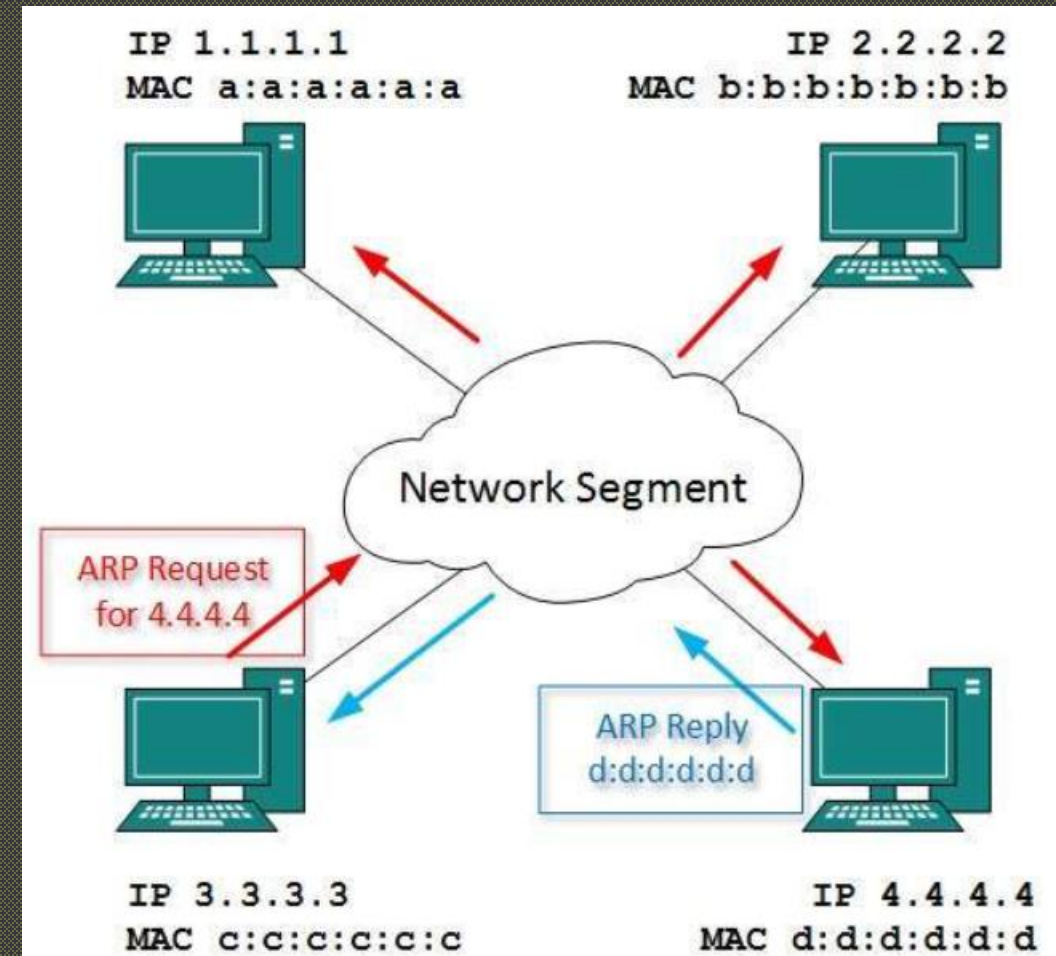
WebRTC – Web Real Time Communications

- Telecommunications Signaling (SS7 over IP):
 - *Example:* The backbone of modern cellular networks (4G LTE, 5G). It is used to set up and manage calls and text messages between your phone and the network towers. Its reliability and multihoming are critical here.
- WebRTC:
 - *Example:* The technology behind browser-based video conferencing and voice calls. SCTP is used as a data channel to reliably transmit non-audio/video data (like chat messages, file transfers) alongside the media streams.
- High-Reliability Services:
 - *Example:* Critical financial trading systems or air traffic control systems where a dropped connection or data loss is not an option. Multihoming ensures the connection stays alive, and multi-streaming ensures timely delivery of priority data.

Feature	SCTP (The Modern Hybrid)	TCP (The Reliable Stream)	UDP (The Fast Datagram)
Connection	Connection-oriented (Association)	Connection-oriented	Connectionless
Reliability	Reliable	Reliable	Unreliable
Data Unit	Messages (Chunks)	Stream of Bytes	Messages (Datagrams)
Key Feature	Multi-Streaming & Multihoming	Strong reliability & flow control	Low overhead & speed
Best For	VoIP, signaling, critical data	Web, email, file transfer	Video streaming, gaming, DNS

5.4 INTERNETWORK LAYER PROTOCOLS

- Internetwork Layer (Core of TCP/IP Suite)
- Main place where network administrators configure settings.
- Runs the IP (Internet Protocol).
- Considered the “heart” of TCP/IP, because:
 - IP addressing happens here.
 - Routing is managed here.
 - Errors in configuration often occur at this layer.
- Without it, the Internet and WWW wouldn't exist.

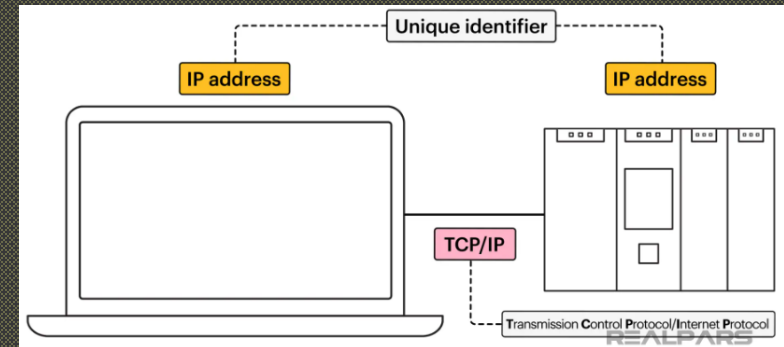


https://www.tutorialspoint.com/data_communication_computer_network/network_layer_protocols.htm

Four Main Tasks of the Internetwork Layer

1. Defines and Verifies IP Addresses

- Each device using TCP/IP gets an IP address (unique identifier).
- Two main purposes of IP address:
 - Host identification → identifies the specific device (like a student number).
 - Network identification → identifies the network the device belongs to (like the campus).
- Structure:
 - Network ID + Host ID (similar to *Area Code* + *Phone Number*).
- Function:
 - When receiving: device checks if the packet's destination IP matches its own.
 - If yes → processes it.
 - If no → discards it.
 - When sending: device adds its own IP as the source address in the packet.
- Example:
 - Computer A (192.168.1.10) sends a message to Computer B (192.168.1.20).
 - B checks the packet → sees that destination = 192.168.1.20 → accepts it.

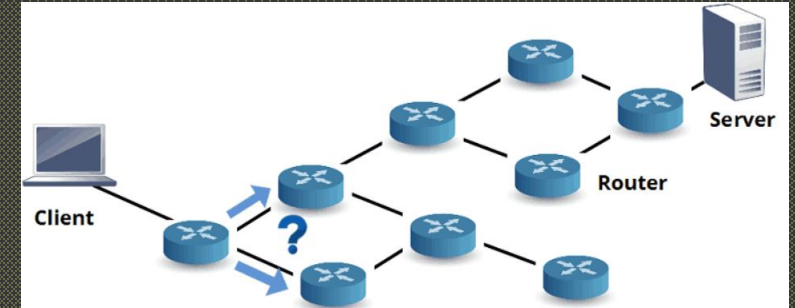


<https://www.realpars.com/blog/static-ip-address-configuration>

5.4 INTERNETWORK LAYER PROTOCOLS

2. Routes Packets through an Internetwork

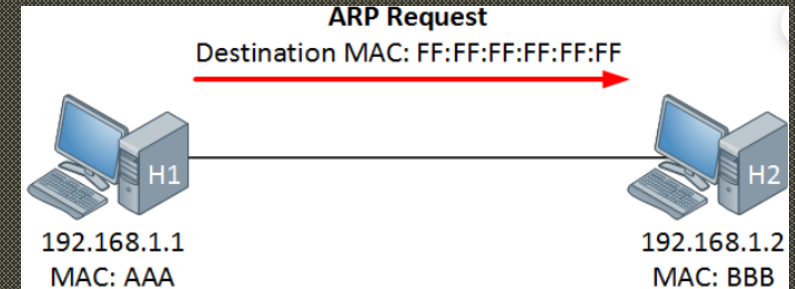
- Determines the best path for data to travel across networks.
- Works similar to a road system:
 - Some routes are fast, some congested, some blocked.
- Routers operate here:
 - Use routing tables and network IDs to choose paths.
 - If one path fails, they find alternatives.
- Example:
- Sending an email from Toronto to Tokyo → packet may travel via multiple routers in the U.S. or Europe before reaching Japan, depending on which path is most efficient.



<https://www.draytek.com/support/knowledge-base/5765>

3. Resolves MAC Addresses from IP Addresses

- Data must include both:
 - Logical address (IP) → works across networks.
 - Physical address (MAC) → unique to the device's network card.
- Uses ARP (Address Resolution Protocol) to find a device's MAC address from its IP.
- Example:
- If Computer A knows Computer B's IP (192.168.1.20), ARP finds B's MAC (e.g., 00:1A:2B:3C:4D:5E) so the packet can actually reach B on the local network.



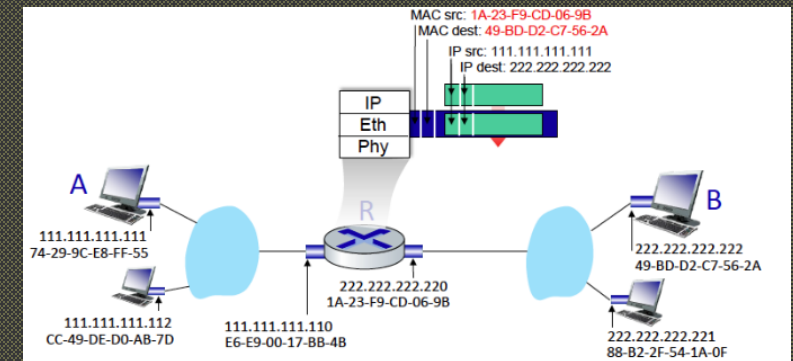
<https://networklessons.com/ip-services/arp-address-resolution-protocol-explained>

4. Delivers Packets Efficiently

- Focuses on speed and low overhead.
- Does not provide:
 - Flow control, error recovery, reassembly, or acknowledgment.
- This is called connectionless communication.
- Relies on upper layers (Transport & Application) for reliability.
 - Example: TCP (Transport Layer) adds acknowledgments → connection-oriented.
 - Example: IP just sends packets without confirmation → connectionless.
- Analogy:
 - Sending a regular letter (IP): no guarantee of delivery.
 - Sending certified mail (TCP): delivery is tracked and confirmed.

Summary

- Internetwork Layer = IP + Routing → essential for internet communication.
- Handles: IP addressing, routing, ARP (MAC resolution), and efficient delivery.
- Errors here cause most network issues.
- Connectionless by design → efficiency first, reliability comes from upper layers.



<https://stevengong.co/notes/ECE358>

5.4 INTERNETWORK LAYER PROTOCOLS

ARP PROTOCOL

Address Resolution Protocol (ARP)

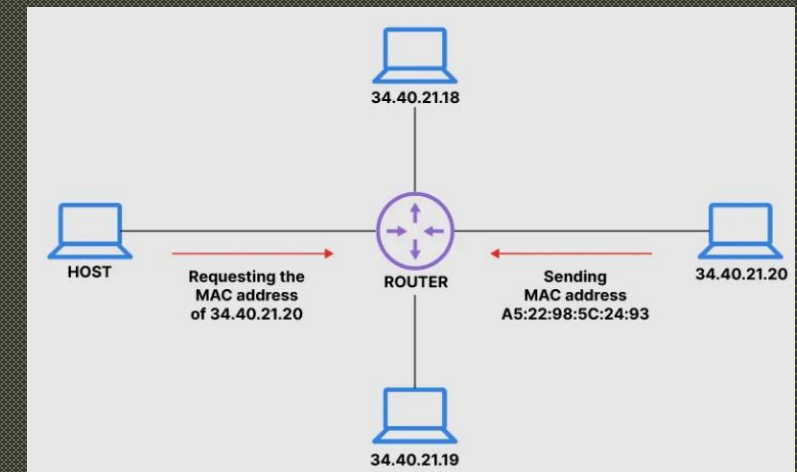
- Maps a logical address (IP) to a physical address (MAC).
- Works inside a local broadcast domain (Layer 2 & 3 interaction).
- Why needed:
 - IP addresses identify devices logically (like a street address).
 - MAC addresses identify devices physically on the local network (like a house number).
 - To send data over Ethernet, the MAC address is required.

Think of sending a letter in an apartment building:

- IP address = Apartment number (logical).
- MAC address = Actual person's nameplate on the door (physical).
- ARP = Asking, "Who lives in Apartment 101?"
- Once you learn the name, you don't need to ask again until you forget (ARP cache).

How ARP Works

- Sending a request (when MAC unknown):
 - Computer A wants to talk to Computer B but only knows B's IP.
 - A sends an ARP broadcast: "Who has IP x.x.x.x? Tell me your MAC."
- Receiving a reply:
 - The device with that IP (B) replies with an ARP reply containing its MAC.
 - Now A can construct the Ethernet frame (Layer 2 header) and send the data.
- Caching results:
 - To avoid constant ARP requests, devices store IP-to-MAC mappings in an ARP cache (RAM).
 - Cache entries are temporary (few minutes).
 - Prevents excessive broadcasts and saves bandwidth.



<https://www.fortinet.com/resources/cyberglossary/what-is-arp#>

5.4 INTERNETWORK LAYER PROTOCOLS

ARP PROTOCOL

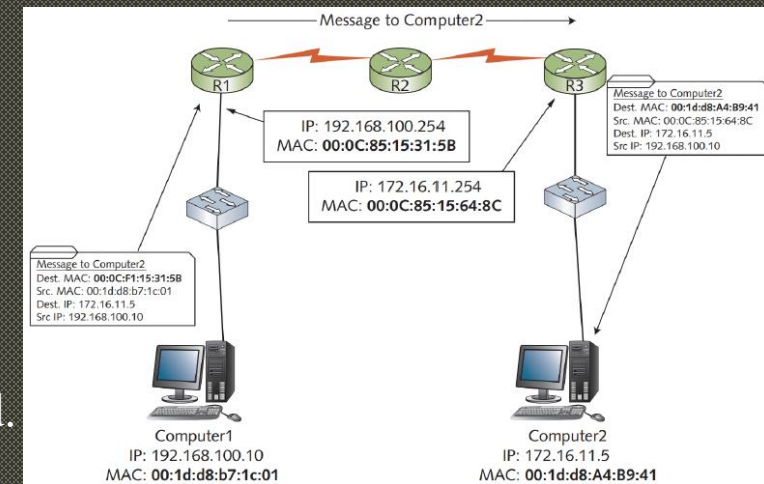
ARP Across Networks

- ARP only works within a local network (broadcast domain).
- Routers do not forward ARP broadcasts → they separate broadcast domains.
- If the destination device is on another network:
 - The source device uses ARP to find the MAC of its default gateway (router).
 - The packet is sent to the router.
 - The router forwards the packet toward the destination network.
 - At the destination network, the final router uses ARP to find the destination host's MAC.

Example:

- Computer A (192.168.100.10) wants to send to Computer B (172.16.11.5).
 - A uses ARP to get the MAC of its router (default gateway).
 - Router forwards the packet across networks.
 - At the destination LAN, the router uses ARP again to find Computer B's MAC.
 - Final delivery happens.
- Why ARP Cache Matters
 - Without caching: Every packet would trigger an ARP broadcast → network overload.
 - With caching: Each IP-MAC pair is learned once, reused until it expires.
 - Entries expire to avoid stale info (e.g., if a NIC is replaced or IP reassigned).

- ARP resolves IP → MAC within a local network.
- Uses broadcast requests and unicast replies.
- Stores results in ARP cache to reduce overhead.
- Routers act as boundaries (do not forward ARP broadcasts).
- Essential for packet delivery across LANs and via gateways to other networks.

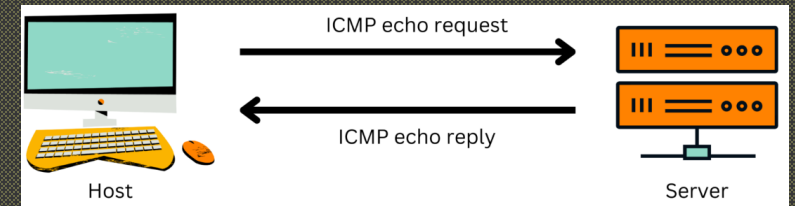


5.4 INTERNETWORK LAYER PROTOCOLS

ICMP PROTOCOL

Internet Control Message Protocol (ICMP)

- Encapsulation: ICMP is carried inside IP packets (wrapped with an IP header).
- Network Diagnostics: Helps admins test connectivity and troubleshoot routing problems.
- Error Reporting: Notifies senders about issues (e.g., unreachable network, packet expired).
- Path Discovery: Allows mapping of network routes.
- Think of ICMP like the postal service notifications:
 - Normal IP packet = the letter you send.
 - ICMP = the message you get back if:
 - The address is wrong (“Destination unreachable”).
 - The mail took too long and was discarded (“Time exceeded”).
 - The recipient confirms they got your letter (Echo Reply).
- Ping (ICMP Echo & Echo Reply)
 - Sends an ICMP Echo Request to a target host.
 - Target replies with an ICMP Echo Reply.
 - Measures:
 - If the host is reachable.
 - Round-trip time (latency).
 - Example: ping google.com → checks if Google’s servers are responding and how fast.
- Traceroute (Tracert on Windows)
 - Uses ICMP to discover the path (hops) a packet takes through routers.
 - Works by sending ICMP Echo packets with increasing TTL (Time To Live) values.
 - Each router that drops the packet when TTL = 0 responds with a TTL Expired message.
 - The sender records the router’s IP address and repeats until it reaches the destination.
 - Example: tracert 8.8.8.8 → shows the sequence of routers between your computer and Google’s DNS server.



<https://www.cloudns.net/blog/what-is-icmp-internet-service-message-protocol/>

5.4 INTERNETWORK LAYER PROTOCOLS

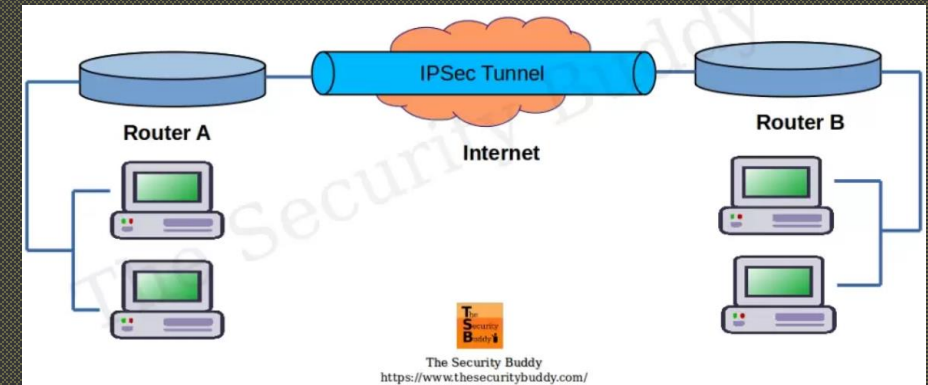
IP_{SEC} PROTOCOL

- Internet Protocol Security is a set of rules and tools that make sure data traveling over the internet is safe and trustworthy.
- It works directly at the IP layer, so it can protect almost any type of internet traffic.
- It is optional in IPv4 but built-in to IPv6.

How IPsec Provides Security

- Authentication
 - Think of this like a passport check for computers.
 - Each computer must prove who it is before sending or accepting data.
 - This can be done with:
 - Passwords
 - Certificates (like digital ID cards)
 - Secret keys shared in advance
 - This prevents attackers from pretending to be a trusted computer.
- Encryption
 - This is like locking your message in a safe before sending it.
 - Even if a hacker intercepts the packet, it will look like scrambled nonsense.
 - Common algorithms: AES (modern, secure) or 3DES (older, less used now).
 - This ensures confidentiality - only the intended receiver can read it.

- Think of IP packets as postcards.
 - Normally, anyone can read them while they travel.
- With IPsec, postcards are put in a sealed, locked envelope that only the sender and receiver can open.



5.4 INTERNETWORK LAYER PROTOCOLS

IP_{SEC} PROTOCOL

Modes of Operation

- Transport Mode:
 - Encrypts only the data portion of the packet, not the IP header.
 - Used for end-to-end communication between two hosts.
- Tunnel Mode:
 - Encrypts the entire packet (header + data), then adds a new IP header.
 - Commonly used for VPNs (Virtual Private Networks).

Advantages of IPsec

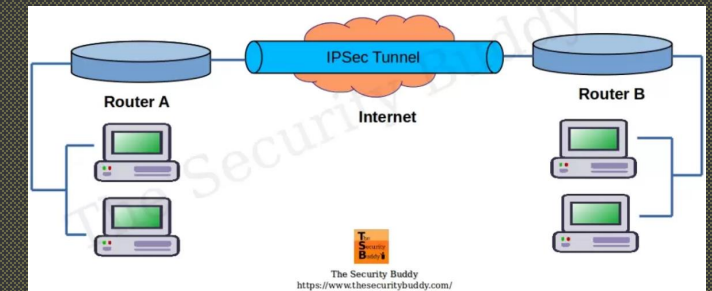
- Data confidentiality → Prevents eavesdropping.
- Data integrity → Ensures data has not been altered in transit.
- Authentication → Confirms sender/receiver identities.
- Protection against replay attacks → Stops attackers from resending captured packets.

Limitations

- Resource intensive: Uses extra CPU and memory for encryption/decryption.
- Not always necessary: Best used only for sensitive communications or in high-risk environments.

Examples of Use

- VPN connections: Secure remote access to corporate networks.
- Secure server-to-server communication: Between data centers or branch offices.
- Protecting sensitive apps: Such as financial transactions, healthcare data, or government communications.



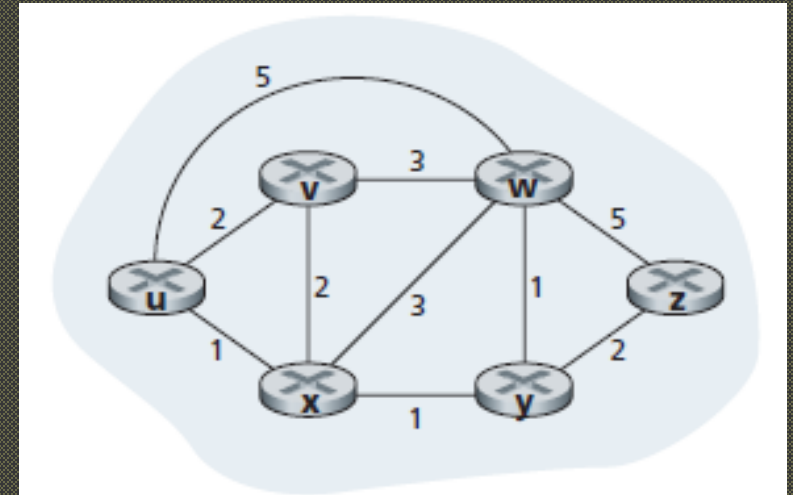
5.4 INTERNETWORK LAYER PROTOCOLS

ROUTING ALGORITHMS

- Goal: Find good paths (routes) for packets from sender → receiver across routers.
- A “good” path = least cost (but can also be influenced by policies).
- Always results in a sequence of routers a packet must travel through.

Graph Model of Networks

- Network: Graph $G = (N, E)$
 - Nodes (N) = routers.
 - Edges (E) = links between routers.
- Each edge has a cost (distance, speed, money, congestion, etc.).
- Path cost = sum of all edge costs along the route.
- Least-cost path = path with the minimum total cost.
 - Example: In a network, the least-cost path from $u \rightarrow w$ could be (u, x, y, w) with total cost = 3.
 - If all edges have the same cost, least-cost path = shortest path (# of hops).



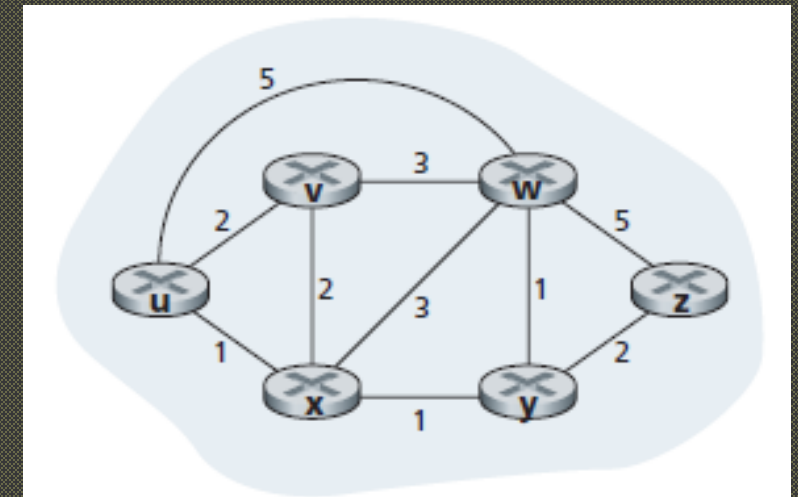
Routing can be restricted by policy; for example, a rule such as “router x should not forward any packets originating from the network owned by organization Z”

5.4

INTERNETWORK LAYER PROTOCOLS ROUTING ALGORITHMS

Routing Algorithms Classification

- Centralized vs. Decentralized
 - Centralized:
 - Knows everything about the whole network (like having a full map).
 - Example: It is like you figuring out all paths on a map with your brain.
 - Type: Link-State (LS).
 - Decentralized:
 - Each router only knows about its nearby neighbors and their costs.
 - They share info step-by-step until they figure out the best routes.
 - Type: Distance-Vector (DV).
- Static vs. Dynamic
 - Static: Paths change rarely (manual updates).
 - Dynamic: Paths update automatically as network changes.
 - More flexible but can cause issues like loops or oscillations.
- Load-sensitive vs. Load-insensitive
 - Load-sensitive: Link costs change with congestion (avoids busy paths).
 - Example: If a link is overloaded, its cost increases → routers find alternative routes.
 - Load-insensitive: Link costs fixed regardless of congestion.
 - Modern Internet protocols (RIP, OSPF, BGP) use this.



5.4 INTERNETWORK LAYER PROTOCOLS

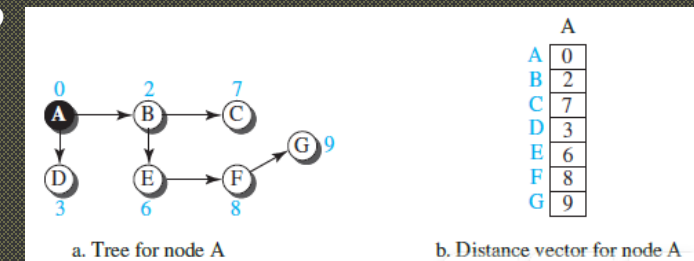
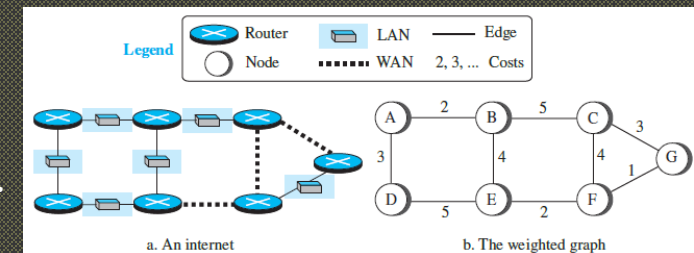
DISTANCE VECTOR ROUTING

The Idea (Bellman-Ford Equation)

Imagine routers are people in a neighborhood who only talk to their next-door neighbors. They pass notes to each other saying, "Here are the distances to all the houses I know about." By sharing this info, everyone gradually learns the shortest way to every house in the neighborhood.

How It Works

- Start Small: Each router only knows its immediate neighbors and the cost to reach them. It knows nothing about the rest of the network.
- Share "Distance Vectors": Each router makes a list (a "vector") of its known distances to every destination and shares this list with its neighbors.
- Update and Learn: When a router gets a list from a neighbor, it does this simple math:
• New Distance = (My cost to Neighbor) + (Neighbor's cost to Destination)
- If this new distance is shorter than its old one, it updates its own list.
- Spread the News: If a router updates its list, it immediately shares the new list with all its neighbors. This process continues until everyone has the same, best information.



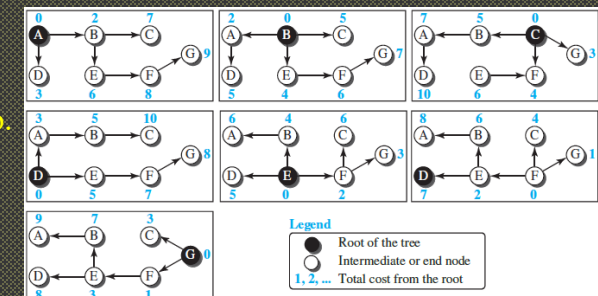
In a Nutshell

What it is: A simple routing method where routers only talk to neighbors to learn the network map.

How it works: They share distance lists and use simple math to find the shortest path.

The Catch: It can create loops when paths fail.

The Solution: Use rules like Split Horizon and Poisoned Reverse to stop the loops.



5.4 INTERNETWORK LAYER PROTOCOLS

DISTANCE VECTOR ROUTING

The Main Problem: Bad News Travels Slowly

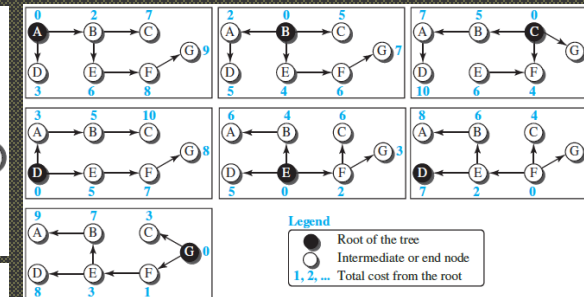
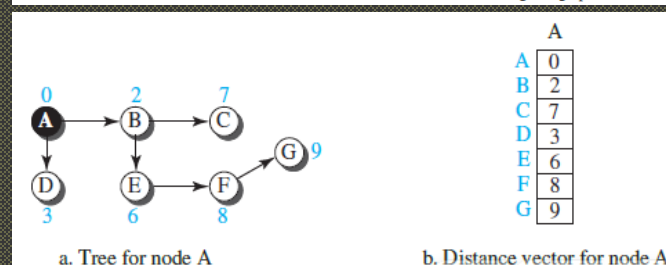
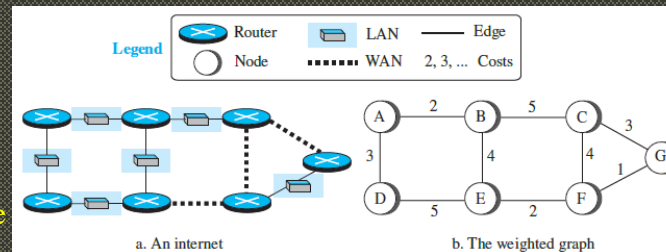
- "Count to Infinity": If a connection breaks, the news of the "broken road" spreads very, very slowly.
- The Result: Routers can get confused and think a broken path is still good, causing data to get stuck in a routing loop (like a package going back and forth between two post offices forever).

To prevent these loops, routers use two main tricks:

- Split Horizon: The rule is, "Do not tell a neighbor about a route you learned from them." This prevents them from getting their own information back and getting confused.
- Poisoned Reverse: This is a stronger version. It says, "If you do tell a neighbor about a route you learned from them, lie and tell them it's broken (infinite cost)." This ensures they will never use that route.

Example: RIP

- The Routing Information Protocol (RIP) uses this method.
- It measures distance in "hops" (how many routers it has to pass through).
- To prevent the slow-bad-news problem from going on forever, RIP says any destination more than 15 hops away is considered unreachable.



5.4 INTERNETWORK LAYER PROTOCOLS

LINK STATE ROUTING

The Idea

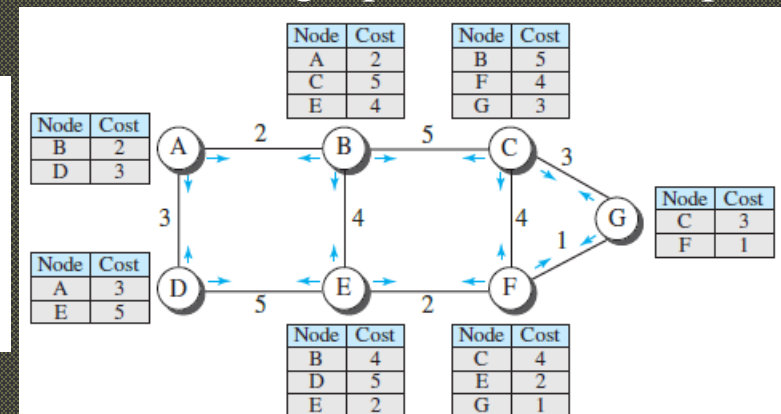
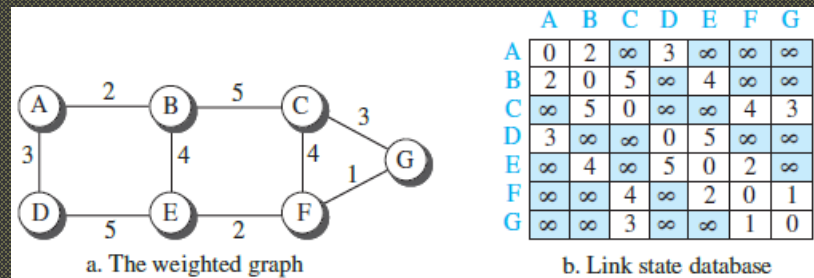
Instead of just talking to neighbors, every router works together to create a complete, identical map of the entire network. Once every router has a copy of the map, they each independently calculate the best routes to every destination.

How It Works

- Discover Neighbors: Each router first figures out who its direct neighbors are and the "cost" to reach each one.
- Shout It Out (Flooding): Each router creates a small message (a "Link-State Packet" or LSP) that lists all its neighbors and the costs to reach them. It then floods this message to the *entire network*.
 - Flooding means: "I tell everyone about my immediate neighbors."
- Build the Map: Every router collects these LSPs from every other router. Once all the messages are in, every router has built the same complete network map, called the Link-State Database (LSDB).
- Calculate the Best Routes: Each router now runs the same computer program (Dijkstra's Algorithm) on this map. This program finds the shortest path from itself to every other router in the network, creating a personal "roadmap" for sending data.

Example: OSPF

- The Open Shortest Path First (OSPF) protocol is the most common real-world example of Link-State routing.



5.4 INTERNETWORK LAYER PROTOCOLS

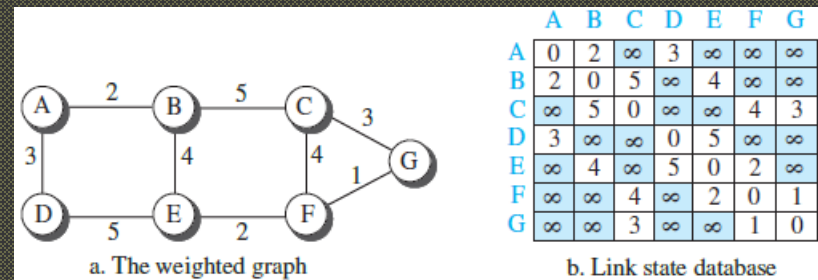
LINK STATE ROUTING

The Main Advantage: Speed and Stability

- **Fast Convergence:** Because everyone has the full map, news about a broken link spreads quickly and reliably. There is no "count to infinity" problem like in Distance-Vector routing.
- **More Reliable:** Each router calculates its own paths from the source of truth (the map), so it's less likely to have routing loops.

The Trade-Off

- This method is more "heavy-duty." It requires more memory to store the full network map and more processing power to run the Dijkstra's algorithm.



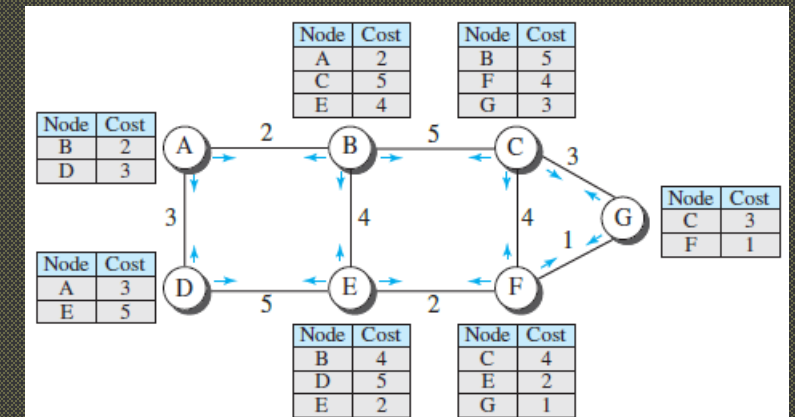
In a Nutshell

What it is: A routing method where every router gets a full, identical map of the network.

How it works: Routers flood info about their local connections to everyone, build a map, and then run Dijkstra's algorithm to find the shortest paths.

The Benefit: Faster, more reliable, and avoids the big problems of Distance-Vector routing.

The Cost: It's more complex and requires more resources from the routers.



5.4 INTERNETWORK LAYER PROTOCOLS

PATH VECTOR ROUTING

The Idea

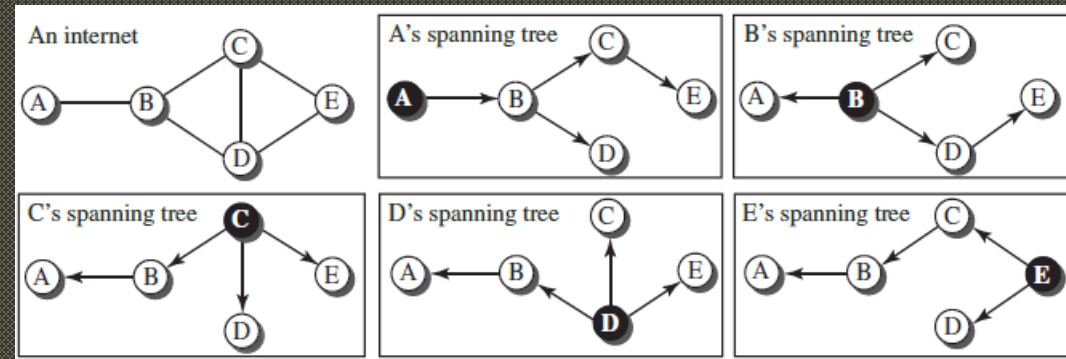
Sometimes, the shortest path is not the best path. Path-Vector Routing lets routers choose routes based on policies (rules and preferences) instead of just distance or cost.

How It Works

- **Share Full Paths, Not Just Costs:** Instead of saying "I am 5 units away from X," a router says, "I get to X through this exact path: Me $\rightarrow$ B $\rightarrow$ C $\rightarrow$ X."
- **Apply Your Policy:** Each router looks at these full paths and picks the best one based on its own rules, like:
 - "Avoid Router D at all costs."
 - "Use the path with the fewest hops."
 - "Only use paths that go through my business partner's network."
- **Prevent Loops Naturally:** If a router sees its own name in a path, it knows that path would create a loop, so it immediately rejects it.

The policy imposed by all sources is to use the minimum number of nodes to reach a destination.

- Nodes A and E select spanning trees that avoid using D as a middle node.
- Node B selects a spanning tree that avoids using C as a middle node..



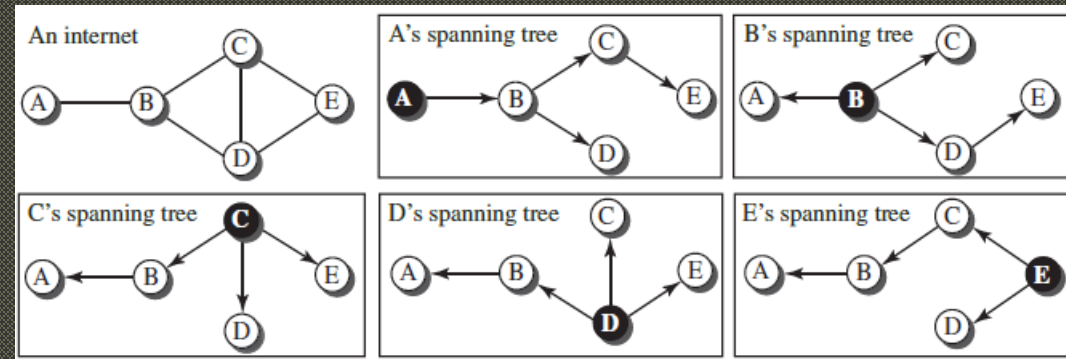
5.4 INTERNETWORK LAYER PROTOCOLS

PATH VECTOR ROUTING

- The Main Advantage: Control
 - This method gives network administrators complete control. They can make routing decisions based on security, business relationships, or reliability, not just on which path is technically the shortest.
- The Trade-Off
 - It is more complex than basic routing methods because it must track and analyze full paths instead of simple numbers.

In a Nutshell

- What it is: A routing method that shares entire paths instead of just costs.
- How it works: Routers exchange lists of paths and use policy rules to choose the best one.
- The Benefit: Gives you control to avoid certain networks or follow business rules.
- Example: This is how the Internet works between major providers (using Border Gateway Protocol - BGP).



5.4 INTERNETWORK LAYER PROTOCOLS

DV – LS – PV ROUTING

Feature	Distance Vector (DV)	Link State (LS)	Path Vector (PV)
Routing Decision Basis	Distance (cost/hops) from self to destination	Global network topology (link states)	Entire path (sequence of nodes/Autonomous Systems)
Information Exchanged	Distance vectors (cost to each destination)	Link-state advertisements (LSAs)	Path vectors (full path lists)
Knowledge of Topology	Each router knows only neighbor costs	Each router has complete network map	Each router knows paths advertised by neighbors
Update Mechanism	Periodic updates to neighbors	Triggered flooding of LSAs	Incremental path updates with policy
Algorithm Used	Bellman-Ford	Dijkstra (Shortest Path First)	Modified Bellman-Ford with policies
Loop Prevention	Split horizon, poison reverse, hold-down timers	Sequence numbers, topology map	Explicit path information (no cycles allowed)
Convergence Speed	Slow (count-to-infinity problem)	Fast (SPF recalculation)	Moderate to slow (policy-driven)
Scalability	Poor (small networks)	Good (large domains)	Excellent (Internet-scale)
Policy Support	None	Limited	Strong (can avoid/choose paths based on rules)
Examples	RIP, IGRP	OSPF, IS-IS	BGP

5.4 INTERNETWORK LAYER PROTOCOLS

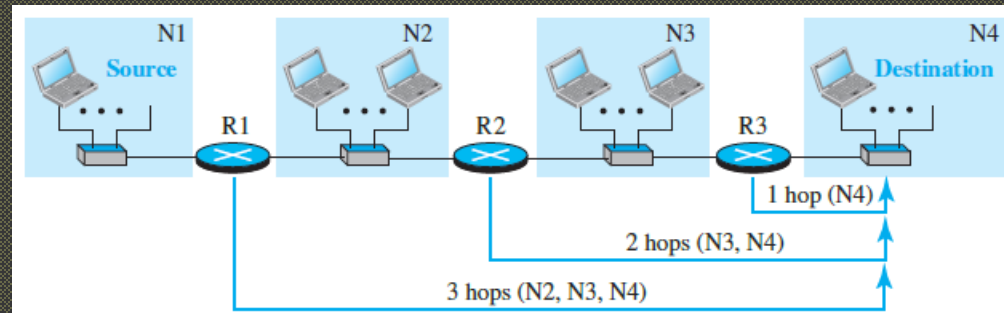
RIP PROTOCOL

The Idea

- RIP is a simple, old-school routing protocol that uses hop count to find the shortest path. It is like giving directions by saying "take the route with the fewest turns."

How It Works

- Count Hops: Every network connection counts as 1 "hop." RIP finds the path with the fewest hops between devices.
- Share Routing Tables: Every 30 seconds, each router tells its neighbors its entire routing table (list of destinations and hop counts).
- Simple Math: When Router A hears from neighbor Router B that "Destination X is 3 hops away," Router A calculates: "If I go through B, Destination X is $3 + 1 = 4$ hops away."
- Small Network Limit: RIP only works in small networks because it considers anything more than 15 hops as unreachable.



5.4 INTERNETWORK LAYER PROTOCOLS

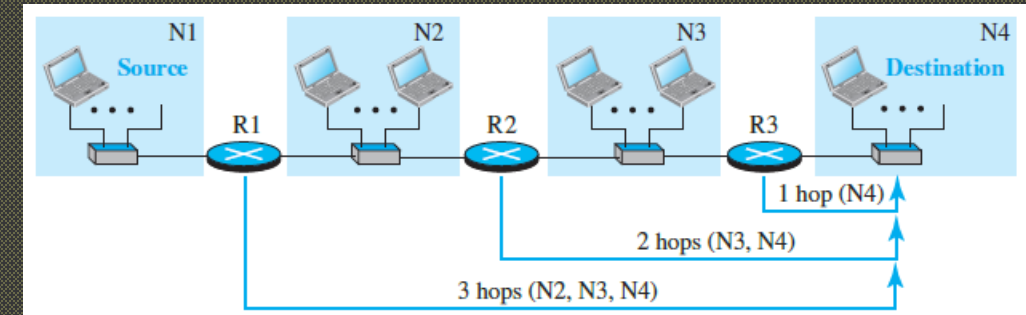
RIP PROTOCOL

Problems

- Slow to React: If a connection breaks, it can take minutes for all routers to realize it.
- "Count to Infinity": Bad news (like a broken link) spreads very slowly through the network.
- Ignores Reality: It only counts hops, ignoring whether a connection is fast or slow.

The Fixes

- Uses tricks like Split Horizon ("don't repeat routes back to where you learned them") and Poison Reverse ("tell neighbors certain routes are broken") to prevent routing loops.



In a Nutshell

- What it is: A simple, hop-count based routing protocol.
- How it works: Routers share their entire routing tables every 30 seconds.
- The Benefit: Extremely simple to set up and understand.
- The Limitations: Only works for small networks (max 15 hops), slow to adapt to changes.
- Status: Largely outdated - was popular in early networks but replaced by better protocols like OSPF in modern networks.

5.4 INTERNETWORK LAYER PROTOCOLS

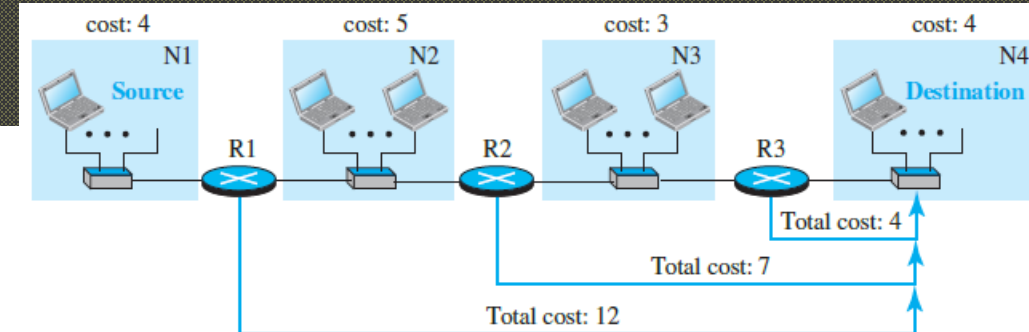
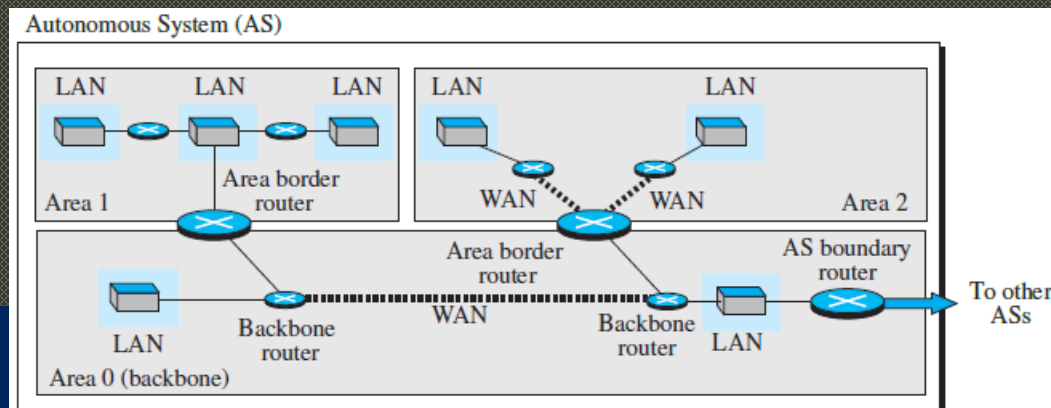
OSPF PROTOCOL

The Idea

- OSPF is a smart routing protocol that gives every router a complete map of the network. Each router then independently calculates the fastest routes using this map, considering factors like speed and reliability, not just distance.

How It Works

- Build the Map: Each router discovers its neighbors and then floods this information to every other router in the area. Everyone ends up with the same complete network map (Link-State Database).
- Calculate Best Paths: Each router runs Dijkstra's algorithm on this map to find the shortest path to every network, creating its own personal "roadmap."
- Handle Large Networks with Areas: For big networks, OSPF divides them into areas connected by a central backbone (Area 0). This prevents the map from getting too big and reduces traffic.



5.4 INTERNETWORK LAYER PROTOCOLS

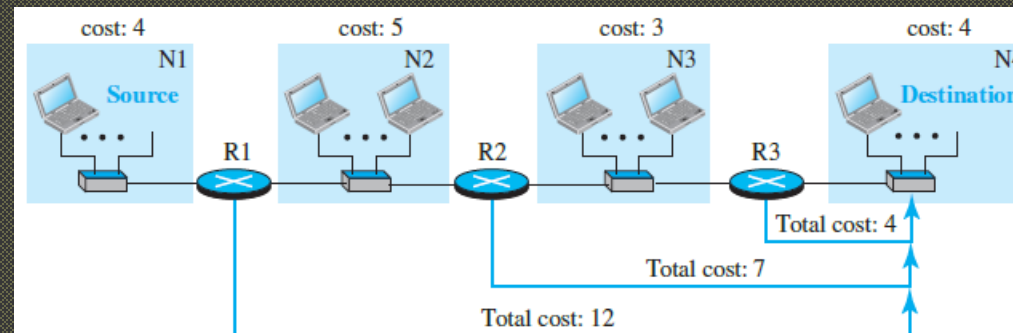
OSPF PROTOCOL

Features

- Smart Metrics: It can consider link speed, delay, and reliability - not just hop count - to find the truly best path.
- Fast Convergence: When a link changes, the news spreads quickly. Routers do not have to slowly figure out that a path is broken.
- Hierarchical Design: The "areas" structure makes it scalable for very large networks.

In a Nutshell

- What it is: An advanced, map-based routing protocol for medium to large networks.
- How it works: Routers flood link information to build a shared map, then run Dijkstra's algorithm to find shortest paths.
- The Benefit: Fast, reliable, and scalable. It understands that the shortest path isn't always the fastest.
- Status: The modern standard for routing within most large organizations and internet service providers.



5.4 INTERNETWORK LAYER PROTOCOLS

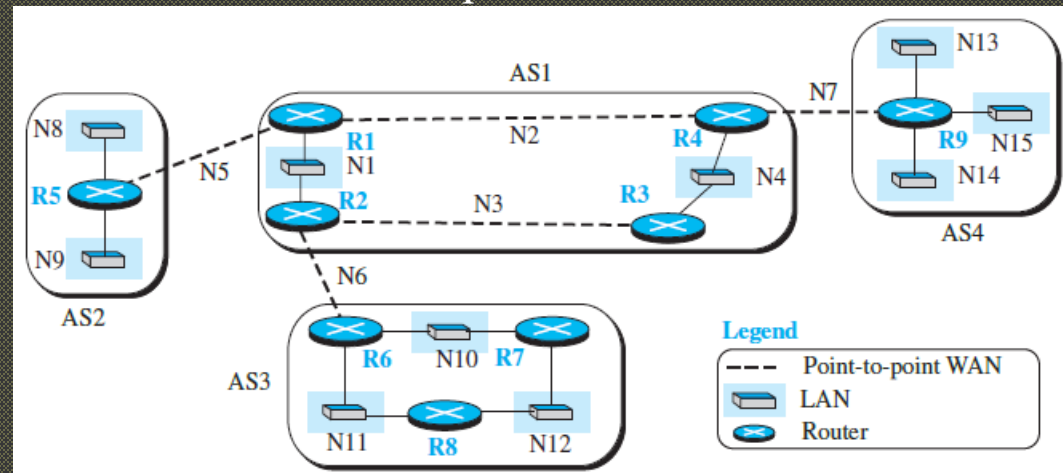
BGP PROTOCOL

The Idea

- Border Gateway Protocol is the postal system of the Internet - it helps different networks (like your ISP, Google, and Netflix) find and talk to each other. Unlike routing inside a single company, BGP focuses on policies and relationships between networks.

How It Works

- Two Types of BGP:
 - eBGP (External): Used between different networks (between your ISP and Google's network)
 - iBGP (Internal): Used within a single network to spread routing information to all routers
- Share Paths, Not Just Distances: Instead of saying "I'm 5 hops away," BGP says "To reach Google, take this path: ISP A → ISP B → Google" and includes business rules.
- Policy-Based Decisions: Networks choose routes based on business relationships:
 - "Prefer routes through our partners"
 - "Avoid routes through competitors"
 - "Use the shortest path through trusted networks"



5.4 INTERNETWORK LAYER PROTOCOLS

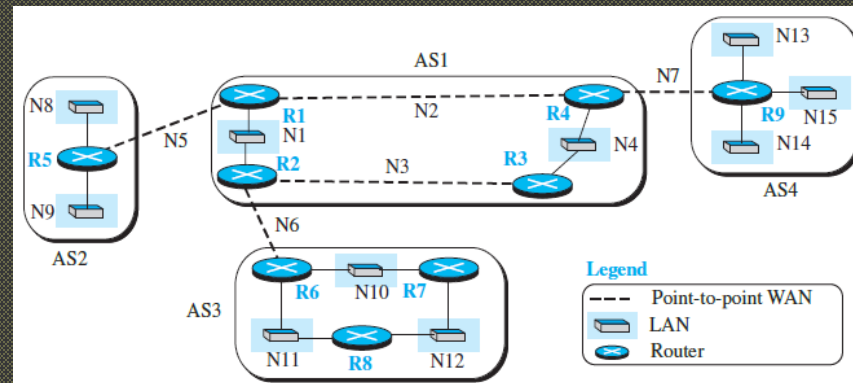
BGP PROTOCOL

Features

- Prevents Loops: By checking if a path already contains their own network (like not sending a letter back to the same post office)
- Scalable: Handles the entire Internet, hundreds of thousands of networks
- Reliable: Uses TCP connections so no messages get lost

In a Nutshell

- What it is: The routing protocol that glues the entire Internet together
- How it works: Networks exchange full paths and apply business policies to choose routes
- The Benefit: Makes the Internet work by connecting all independent networks
- The Trade-off: More complex and slower than internal routing protocols
- Status: The essential backbone of the Internet - without BGP, the Internet would break into disconnected pieces



5.4 INTERNETWORK LAYER PROTOCOLS

RIP – OSPF - BGP PROTOCOLS

Feature	RIP	OSPF	BGP
Protocol Type	Distance Vector	Link State	Path Vector
Metric	Hop Count (max 15)	Cost (based on bandwidth)	Policy-based (AS-PATH, LOCAL-PREF, etc.)
Transport Protocol	UDP (port 520)	Runs directly over IP (protocol 89)	TCP (port 179)
Scope	Intra-domain (small networks)	Intra-domain (large networks)	Inter-domain (between ASes)
Max Network Size	Small (<15 hops)	Large enterprise networks	Global Internet
Update Mechanism	Periodic full table updates (every 30s)	Event-triggered (LSAs)	Incremental updates (on changes)
Convergence Speed	Slow (count-to-infinity problem)	Fast (Dijkstra's algorithm)	Slow (policy-driven decisions)
Loop Prevention	Split Horizon, Poison Reverse	Dijkstra + sequence numbers	AS-PATH attribute
Message Overhead	Low (simple tables)	Moderate (LSAs flood network)	High (path attributes exchanged)
Administrative Use	Simple LANs, legacy systems	Large enterprises, ISPs internally	Internet backbone, ISP-to-ISP
Advantages	Easy to configure, lightweight	Efficient, scalable, fast convergence	Global scale, policy control
Disadvantages	Not scalable, slow convergence	More complex, requires CPU/memory	Complex, slow convergence, high overhead

Summary:

- RIP = Simple but limited.
- OSPF = Scalable and efficient for enterprises.
- BGP = The Internet's backbone, policy-driven, global scale.

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INDUSTRIAL NETWORKING

MENG 4010

Thank you!

Discussions?